

TOR 4. Use the appropriate assessment approach to estimate annual fishing mortality, recruitment and stock biomass (both total and spawning stock) for the time series, and estimate their uncertainty. Compare the time series of these estimates with those from the previously accepted assessment(s). Evaluate a suite of model fit diagnostics e.g., residual patterns, sensitivity analyses, retrospective patterns), and (a) comment on likely causes of problematic issues, and (b), if possible and appropriate, account for those issues when providing scientific advice and evaluate the consequences of any correction(s) applied.

Scallop Research Track Working Group

Northeast Fisheries Science Center, NOAA

April 2025



Overview

- ▶ CASA Overview
 - 1. Stock Boundary
 - 2. Catch and Survey Data
 - 3. Population
 - 4. Growth
 - 5. Natural Mortality
- ▶ Mid-Atlantic Model
- ▶ Georges Bank Closed Model
- ▶ Georges Bank Open Model
- ▶ All-Three Models Combined

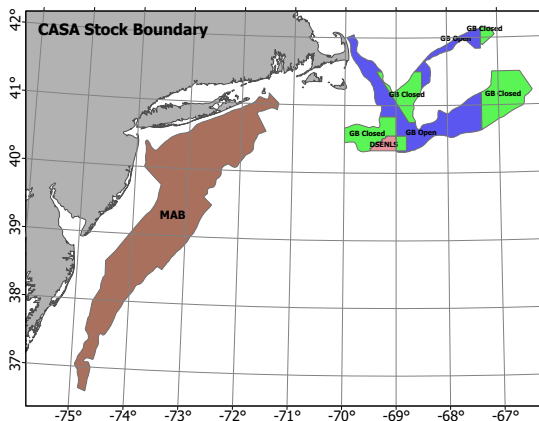
CASA (Catch-At-Size-Analysis)

- ▶ CASA was developed based on Sullivan et al. 1990¹.
- ▶ Size-structured model.
- ▶ Appropriate for sea scallops (abundant commercial and population shell height data, and growth increment data).
- ▶ Parameters are estimated using likelihood methods.
- ▶ Used since 2007 assessment (SARC-45), but with some refinements for each assessment.

¹Sullivan PJ, Lai HL, Gallucci VF (1990) A catch-at-length analysis that incorporates a stochastic model of growth. Can J Fish Aquat Sci 47: 184-198

Stock Boundary

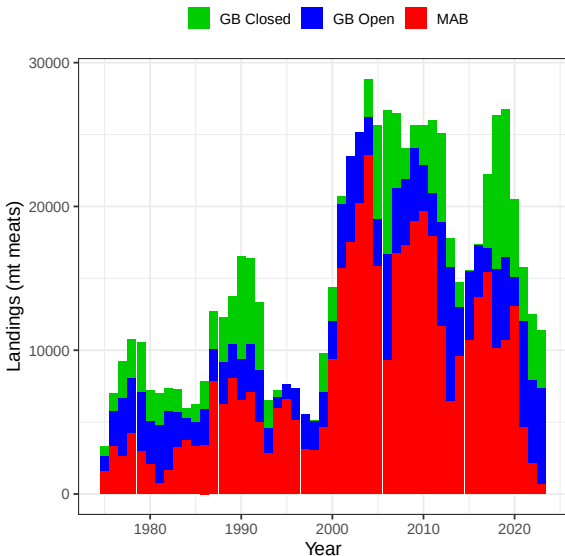
US sea scallop was assessed using three CASA models: Mid-Atlantic, Georges Bank Closed, and Georges Bank Open², excluding scallops in the Deep Southeast Nantucket Lightship (DSENLs).



²Hart DR, Jacobson LD, Tang J (2013) To split or not to split: Assessment of Georges Bank sea scallops in the presence of marine protected areas. *Fish Res* 144: 74–83

Commercial Catch Data Used in CASA

- ▶ CASA modeling year: 1975-2023
- ▶ Commercial landings: 1975-2023
- ▶ Commercial size composition: 1975-1984 from port samples, and 1992-2023 from observers.



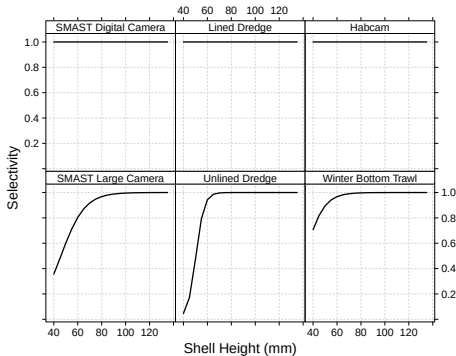
Commercial Catch Data Used in CASA

- ▶ Several fishery selectivity periods were implemented for each stock.
- ▶ Most of the fishery selectivities were modeled logistic, but some domed.
- ▶ Domed selectivities due to fishery targeting intermediate-sized scallops.

Fishery Selectivity Period	Mid-Atlantic	Georges Bank Closed	Georges Bank Open
1	1975-1979 (domed)	1975-1997 (logistic)	1975-1998 (logistic)
2	1980-1997 (logistic)	1998-2017 (logistic)	1999-2004 (logistic)
3	1998-2001 (logistic)	2017 (logistic)	2005-2023 (logistic)
4	2002-2004 (logistic)	2018 (logistic)	-
5	2005-2023 (logistic)	2019 (domed)	-
6	-	2020-2021 (logistic)	-
7	-	2022-2023 (domed)	-

Survey Data Used in CASA

- ▶ CASA models were tuned to 6 surveys.
- ▶ Survey selectivities were fixed in CASA.
- ▶ Beta prior on survey catchability (mean=0.5) was applied to 4 surveys to incorporate knowledge regarding absolute scale from these surveys.



Survey	Mid-Atlantic	Georges Bank Closed and Open	Selectivity Type	Q Prior
NEFSC unlined dredge	✓ (1975, 1977-78)	✓ (1975, 1977)	Logistic	
NEFSC/VIMS lined dredge (1979-2023)	✓	✓	Flat	✓
NEFSC winter bottom trawl (1992-2007)	✓		Logistic	
SMAST large camera (2003-12, 2014-15)	✓	✓	Logistic	✓
SMAST digital camera	✓ (2017, 2019)	✓ (2017, 2019, 2021-23)	Flat	✓
NEFSC/Arnies/WHOI/CFF Habcam	✓ (2012-23)	✓ (2011-19, 2021, 2022)	Flat	✓

Catch and Survey Data in CASA

- ▶ Data $>40\text{mm}$ were used for tuning.
- ▶ Catch and survey time series and size compositions were fitted in CASA with the assumed CVs and effective sample sizes.
- ▶ Effective sample sizes for size composition data were tuned in preliminary model runs so that the median of assumed values used in tuning were similar to expected values based on the goodness of fit³.

³ Butler JL, Jacobson LD, Barnes JT, Moser HC (2003) Biology and population dynamics of cowcod (*Sebastes levis*) in the southern California Bight. Fish Bull 101: 260-280

Population in CASA

- ▶ Population started at 0-5mm, with 5mm size bin.
- ▶ Recruitment was estimated at age 1 (0-35mm), with no assumed stock-recruitment relationship.
- ▶ Population shell height to meat weight conversion used parameters estimated from 2001-23 survey data.
- ▶ Fishery meat weights were estimated using the above parameters, and adjusted based on seasonal anomalies from observer data.

Growth in CASA

- ▶ Growth in CASA was advanced using growth transition matrices estimated from growth increment data.
- ▶ Strong evidence of growth changing over time.
- ▶ Several growth transition matrices were used to represent growth in different time periods.

Growth Period (Slow to Fast)	Year
Mid-Atlantic	
1	1990-99
2	1975-78, 1984-89, 2000-03, 2016-18
3	1979, 1982, 1983, 2004-08, 2014, 2015, 2019, 2020
4	1980, 1981, 2009-13, 2021-23
Georges Bank Closed	
1	1986-97, 2015-23
2	1975-85, 1998-2014
Georges Bank Open	
1	1987-90, 2011-23
2	1975-86, 1991-2010

Natural Mortality in CASA⁴

Option 1: Size-Specific Annual M

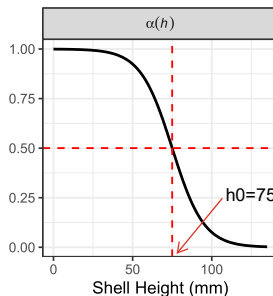
$$M(h, y) = \alpha(h)M_{\text{juv}}(y) + [1 - \alpha(h)]M_{\text{ad}}(y),$$

where juvenile M is $M_{\text{juv}}(y) = \bar{M}_{\text{juv}} + \theta(y)$ and adult M is $M_{\text{ad}}(y) = \bar{M}_{\text{ad}} + \omega(y)$, both can be decomposed into their mean values and annual deviations from the mean. $\alpha(h)$ is a decreasing logistic function by shell height h used to partition natural mortalities between juveniles and adults.

Option 2: Annual M

$$M(y) = \bar{M} + \gamma(y),$$

where $\bar{M}$ is mean natural mortality and $\gamma(y)$ is annual deviations from the mean.

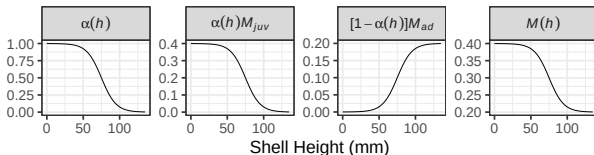


⁴

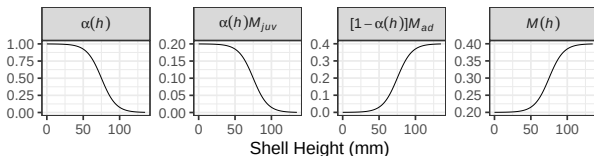
Hart DR, Chang JH (2022) Estimating natural mortality for Atlantic Sea scallops (*Placopecten magellanicus*) using a size-based stock assessment model. Fish Res 254: 106423.

Natural Mortality in CASA

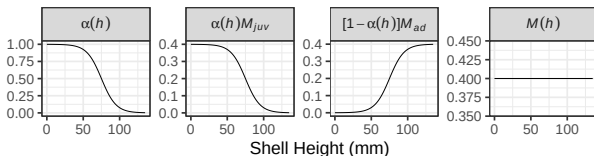
$$M(h, y) = \alpha(h)M_{juv}(y) + [1 - \alpha(h)]M_{ad}(y) \quad M_{juv} = 0.4 \text{ and } M_{ad} = 0.2$$



$$M_{juv} = 0.2 \text{ and } M_{ad} = 0.4$$



$$M_{juv} = M_{ad} = 0.4$$



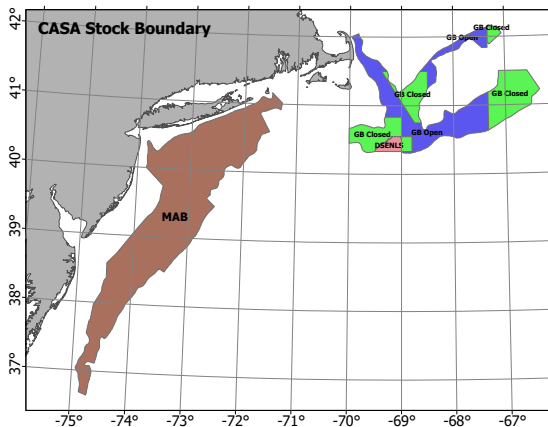
Natural Mortality in CASA

Estimation of natural mortality for each stock:

Natural Mortality Parameters	Mid-Atlantic	Georges Bank Closed	Georges Bank Open
Option 1: $M(h, y) = \alpha(h)M_{juv}(y) + [1 - \alpha(h)]M_{ad}(y)$			
Mean Juvenile M [$\bar{M}_{juv}$]	Fixed at 0.4	-	Fixed at 0.27
Annual Deviation for Juvenile M [$\theta(y)$]	Estimated	-	Estimated
Mean Adult M [$\bar{M}_{ad}$]	Fixed at 0.4	-	Fixed at 0.27
Annual Deviation for Adult M [$\omega(y)$]	Estimated	-	-
Logistic curve [$\alpha(h)$]	Fixed (75mm)	-	Fixed (75mm)
Option 2: $M(y) = \bar{M} + \gamma(y)$			
Mean Total M [$\bar{M}$]	-	Estimated	-
Annual Deviation for Total M [$\gamma(y)$]	-	Estimated	-

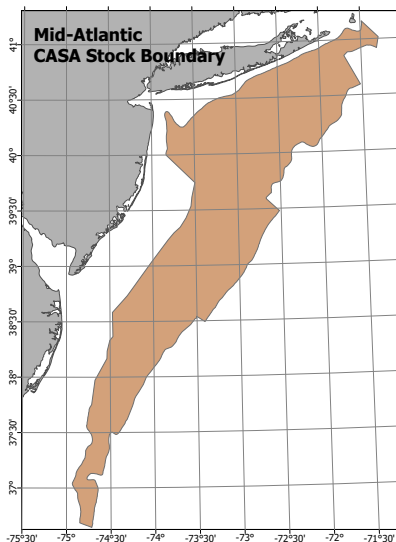
Next: Stock-Specific CASA Models and Results

- ▶ Base case model configuration and model results for the three stocks.
- ▶ Started with MAB, followed by GB closed, lastly GB open models.
- ▶ Retrospective analysis, likelihood profiles, and sensitivity analysis were used to describe uncertainties for each stock.



Mid-Atlantic Base Case Model Configuration

- ▶ Surveys: unlined dredge, lined dredge, winter bottom trawl, SMAST large video camera, SMAST digital camera, and Habcam surveys.
- ▶ Fishery selectivity periods: 1975-1979 (domed), 1980-1997, 1998-2001, 2002-2004, and 2005-2023.
- ▶ Assumed 4 growth periods, with the fastest growth in the 80s and most recent years 2021-2023.



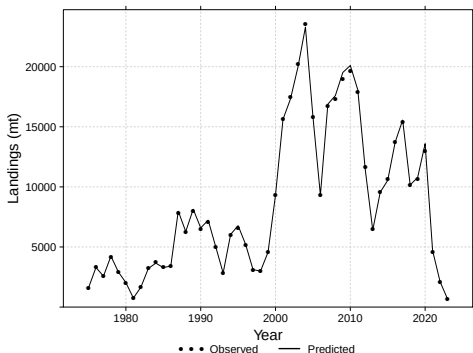
Mid-Atlantic Base Case Model Configuration

- ▶ In SARC 65, mean M was fixed at 0.25 and only estimated juvenile annual deviance.
- ▶ Increase in adult M observed in multiple surveys in recent years.
- ▶ In RT 2025, mean M was increased to 0.4 based on a profile analysis and literature, both juvenile and adult deviance were estimated.

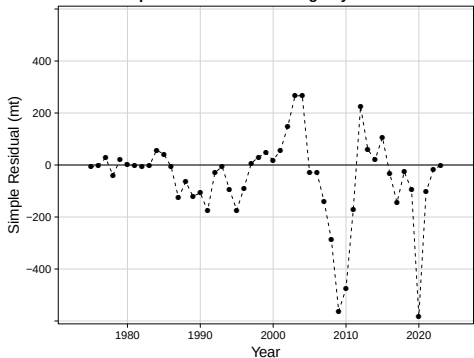
Natural Mortality Parameters	SARC-65	RT 2025
Option 1: $M(h, y) = \alpha(h)M_{\text{juv}}(y) + [1 - \alpha(h)]M_{\text{ad}}(y)$		
Mean Juvenile M [$\bar{M}_{\text{juv}}$]	Fixed at 0.25	Fixed at 0.4
Annual Deviation for Juvenile M [$\theta(y)$]	Estimated	Estimated
Mean Adult M [$\bar{M}_{\text{ad}}$]	Fixed at 0.25	Fixed at 0.4
Annual Deviation for Adult M [$\omega(y)$]	-	Estimated
Logistic curve [$\alpha(h)$]	Fixed (70mm)	Fixed (75mm)
Option 2: $M(y) = \bar{M} + \gamma(y)$		
Mean Total M [$\bar{M}$]	-	-
Annual Deviation for Total M [$\gamma(y)$]	-	-

Mid-Atlantic Base Case: Observed/Predicted Catch

**Mid-Atlantic
Observed And Predicted Landings**

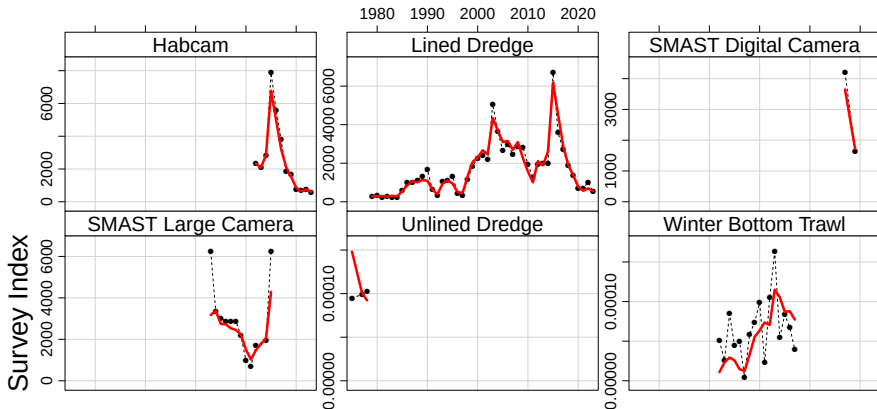


**Mid-Atlantic
Simple Residuals Of Landings By Year**



Mid-Atlantic Base Case: Observed/Predicted Survey Indices

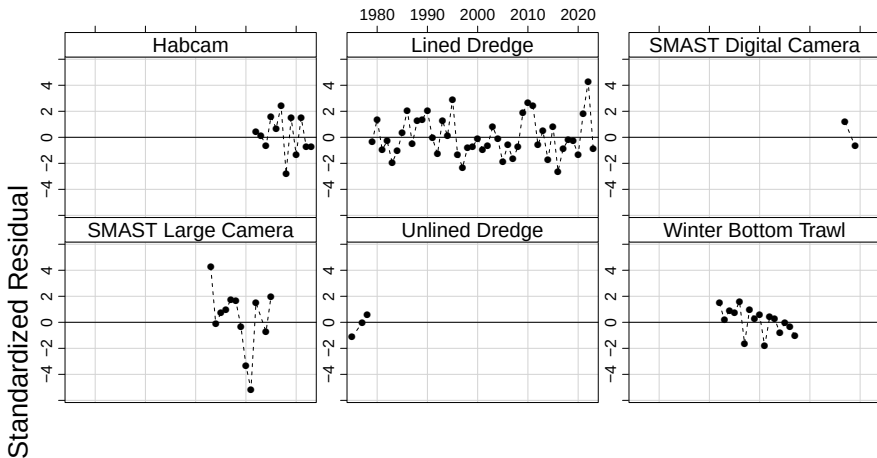
Mid-Atlantic Observed And Predicted Survey Indices By Year



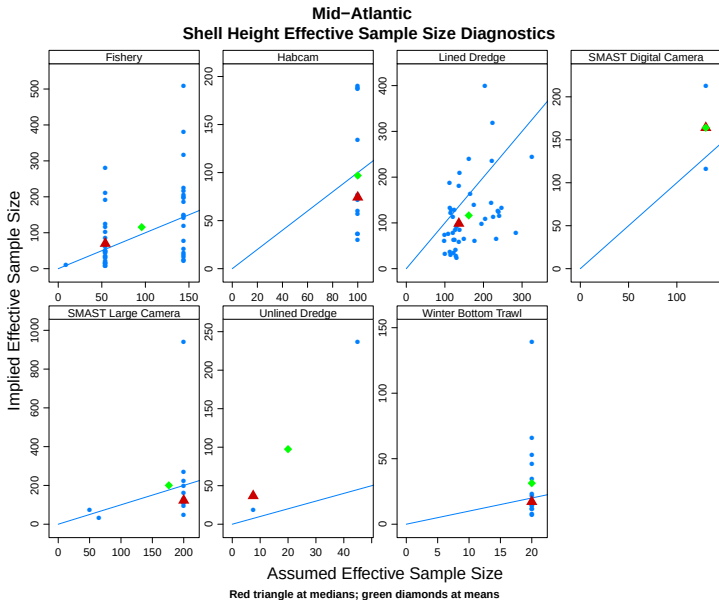
Black: Observed; Red: Predicted

Mid-Atlantic Base Case: Standardized Residuals of Survey Indices

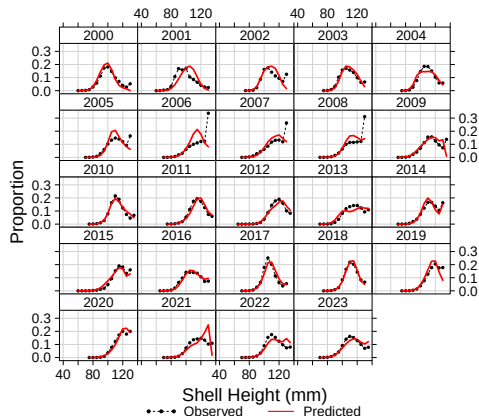
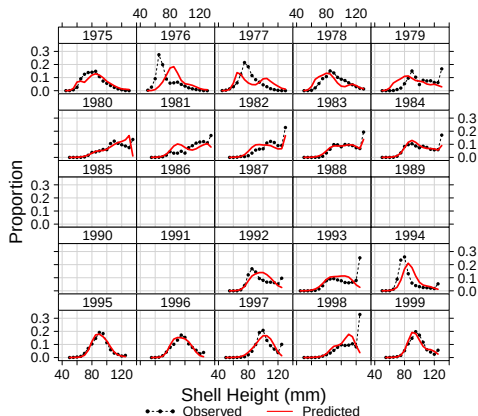
Mid-Atlantic Standardized Residuals Of Survey Indices By Year



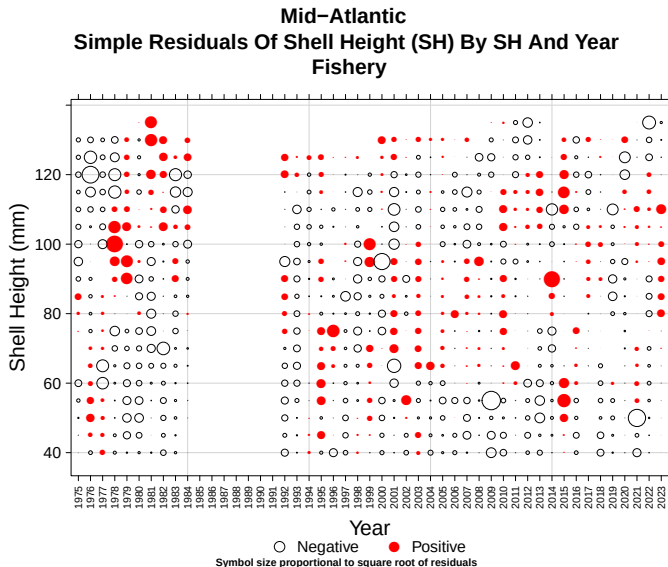
Mid-Atlantic Base Case: Effective Sample Size



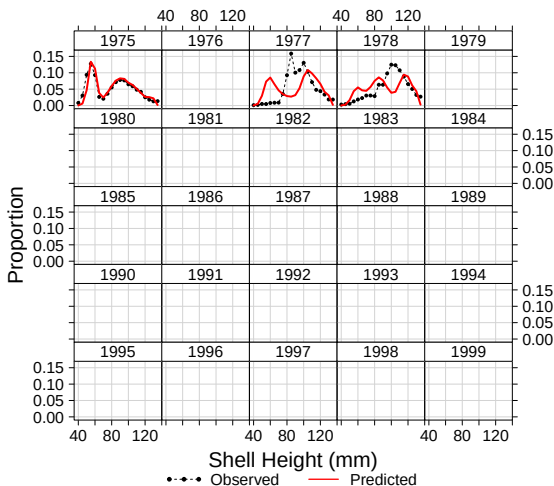
Mid-Atlantic Base Case: Fishery Size Freq.



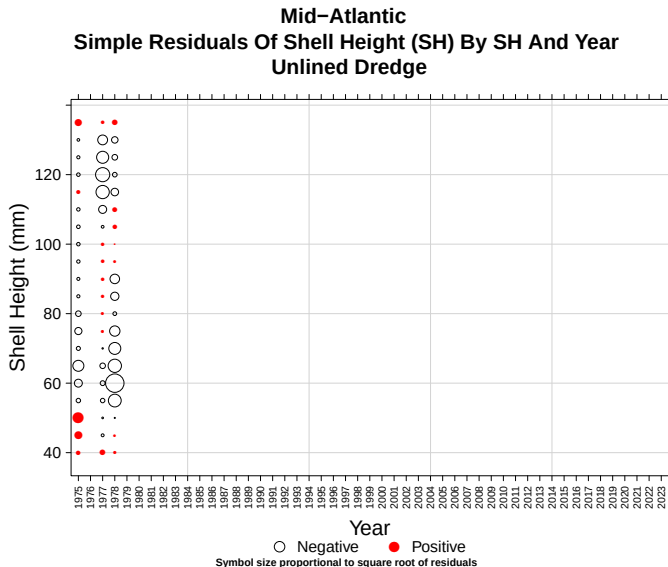
Mid-Atlantic Base Case: Fishery Size Freq. Residuals



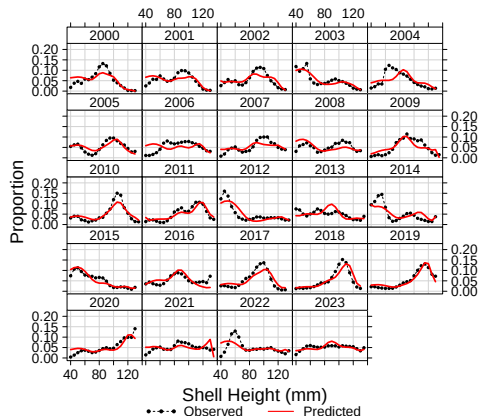
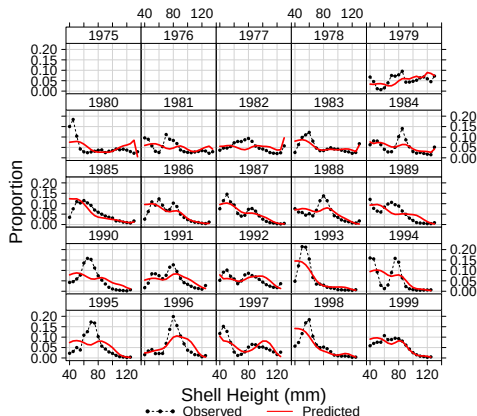
Mid-Atlantic Base Case: Unlined Dredge Size Freq.



Mid-Atlantic Base Case: Unlined Dredge Size Freq. Residuals

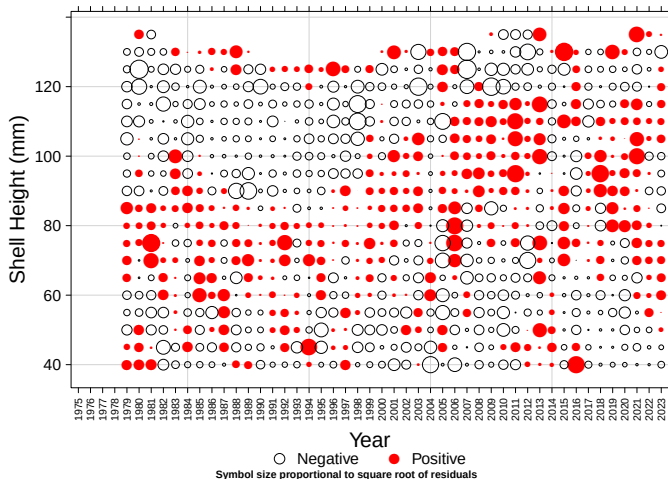


Mid-Atlantic Base Case: Lined Dredge Size Freq.

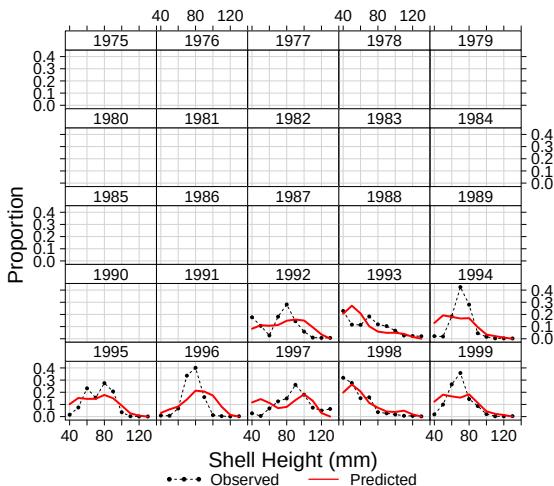


Mid-Atlantic Base Case: Lined Dredge Size Freq. Residuals

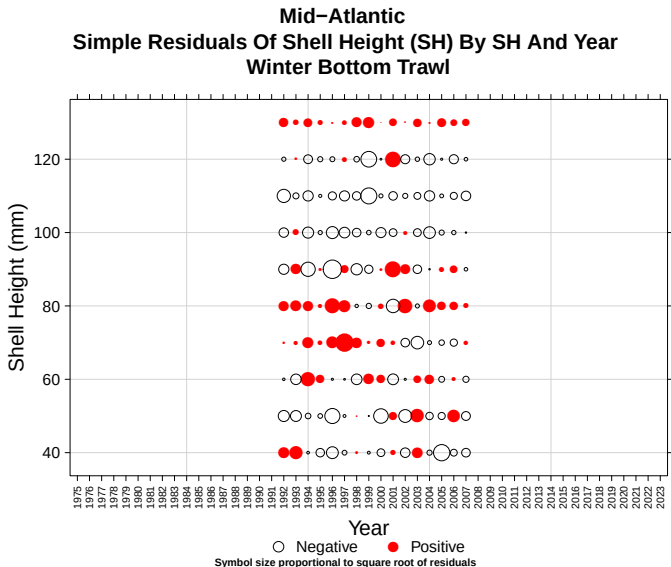
Mid-Atlantic
Simple Residuals Of Shell Height (SH) By SH And Year
Lined Dredge



Mid-Atlantic Base Case: Winter Bottom Trawl Size Freq.

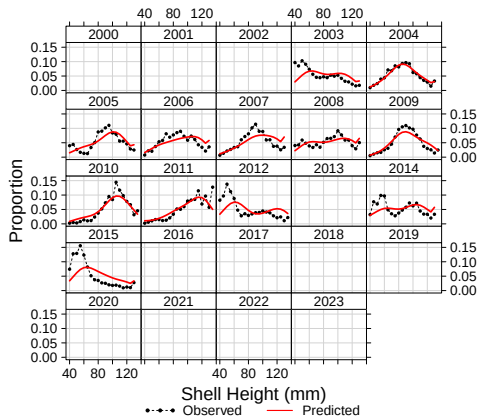


Mid-Atlantic Base Case: Winter Bottom Trawl Size Freq. Residuals

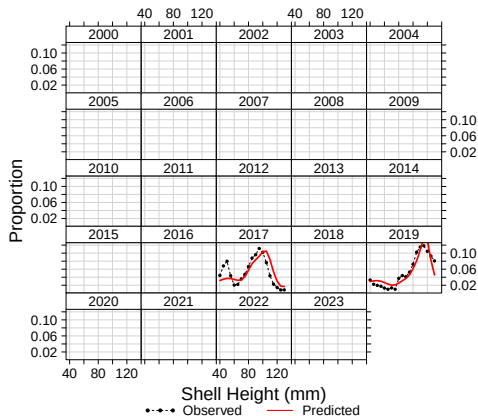


Mid-Atlantic Base Case: SMAST Size Freq.

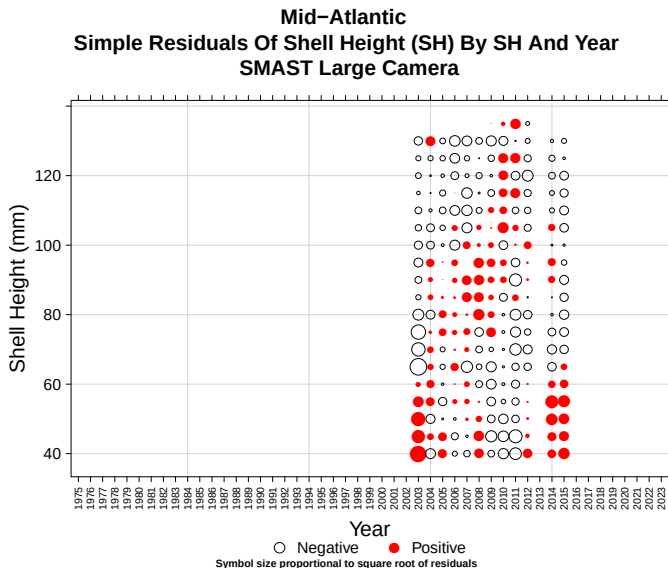
SMAST Large Camera Survey



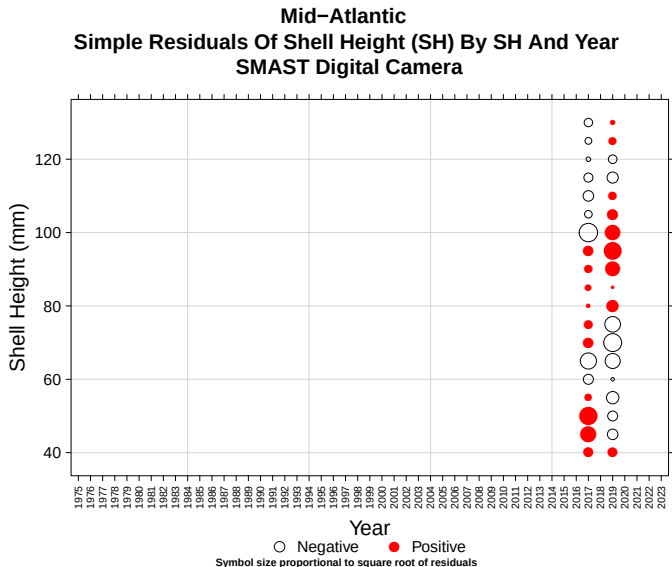
SMAST Digital Camera Survey



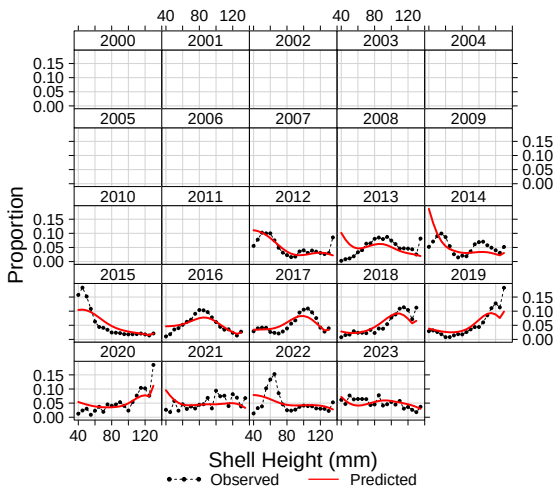
Mid-Atlantic Base Case: SMAST Large Camera Size Freq. Residuals



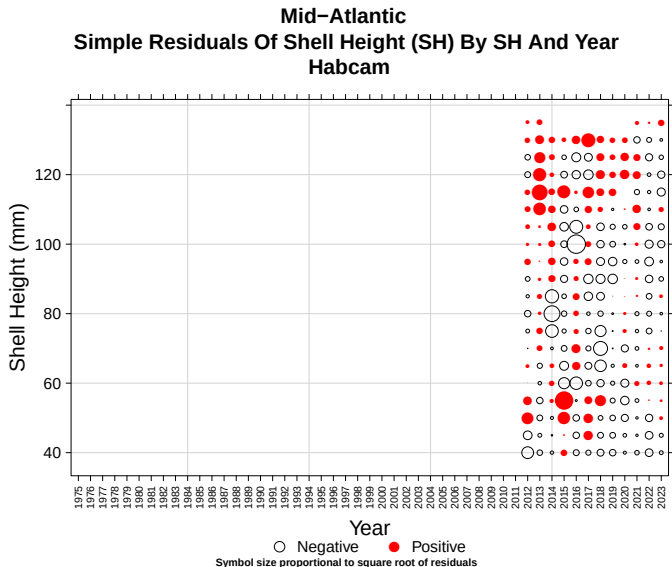
Mid-Atlantic Base Case: SMAST Digital Camera Size Freq. Residuals



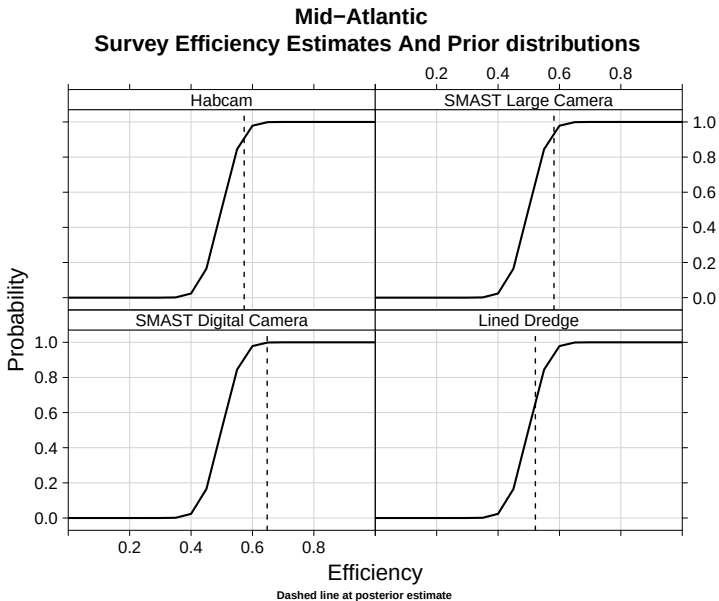
Mid-Atlantic Base Case: Habcam Survey Size Freq.



Mid-Atlantic Base Case: Habcam Size Freq. Residuals

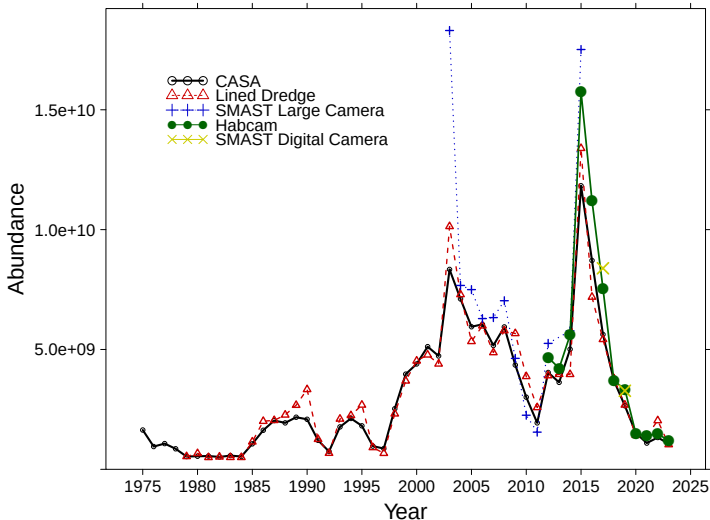


Mid-Atlantic Base Case: Survey Catchability



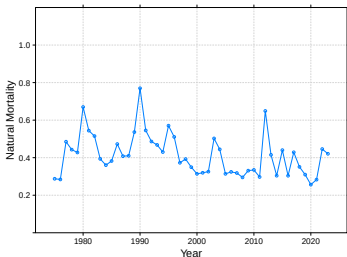
Mid-Atlantic Base Case: Observed and CASA-Estimated Abundance

Mid-Atlantic Survey And Model Estimated Abundance By Year

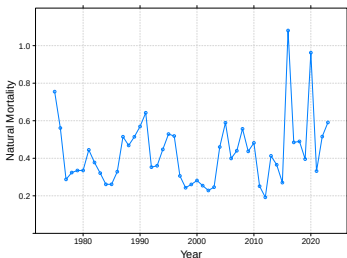


Mid-Atlantic Base Case: Natural Mortality by Size and Year

Mid-Atlantic
Natural Mortality At Smallest Size

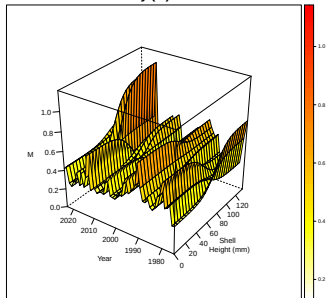


Mid-Atlantic
Natural Mortality At Shell Height 135 mm

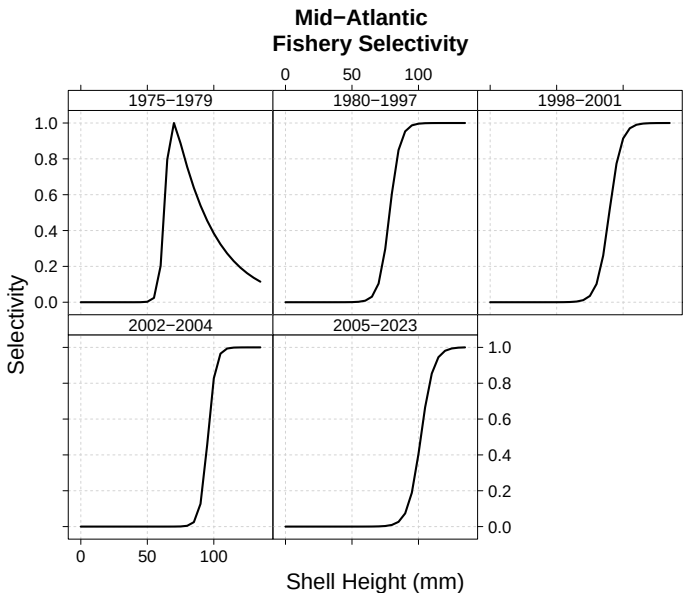


- ▶ Mean M is 0.42 across all years and sizes for this assessment, while the mean M was 0.26 for SARC-65.
- ▶ Adult M (75-135mm) was 0.4 before 2016 and 0.58 from 2016 to 2023.
- ▶ Elevated adult M for these years is consistent with the declines observed in multiple surveys.

Mid-Atlantic
Natural Mortality (M) From All Sources

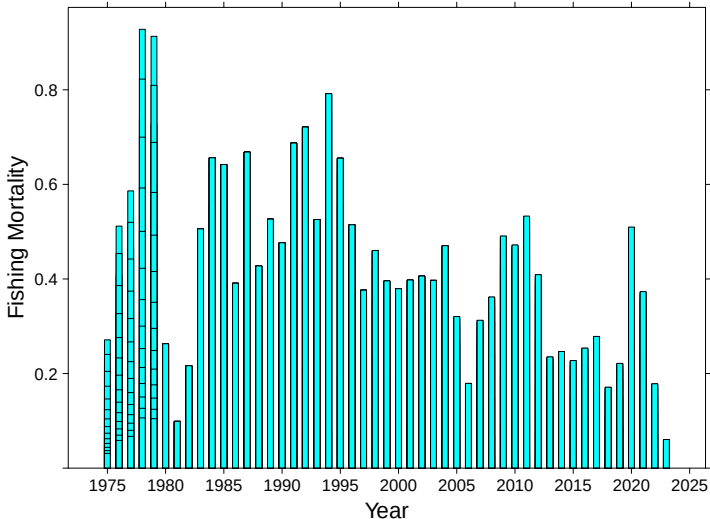


Mid-Atlantic Base Case: Fishery Selectivity

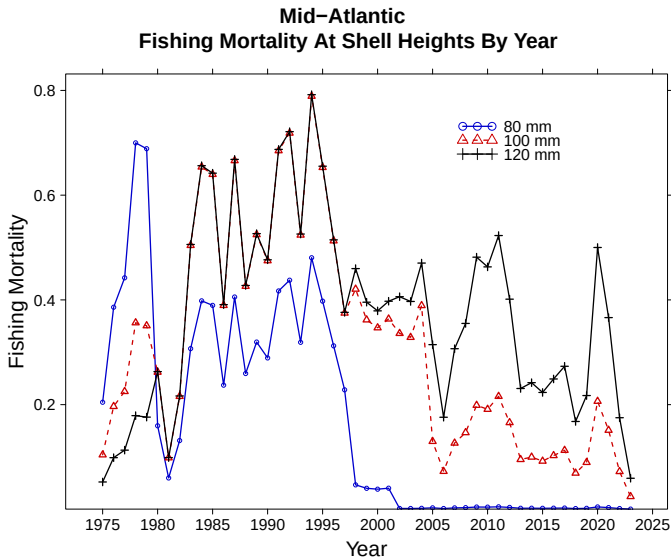


Mid-Atlantic Base Case: Estimated Fully Recruited Fishing Mortality

**Mid-Atlantic
Fully Recruited Fishing Mortality By Year**

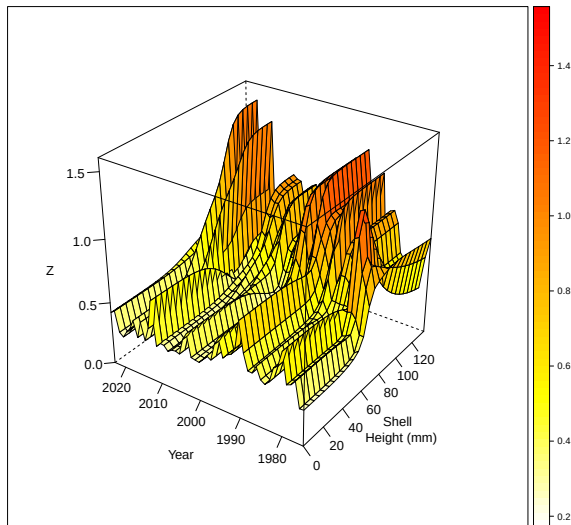


Mid-Atlantic Base Case: Estimated Fishing Mortality By Size



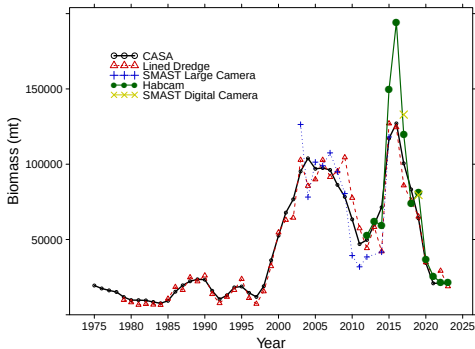
Mid-Atlantic Base Case: Total Mortality

**Mid-Atlantic
Total Mortality (Z) From All Sources**

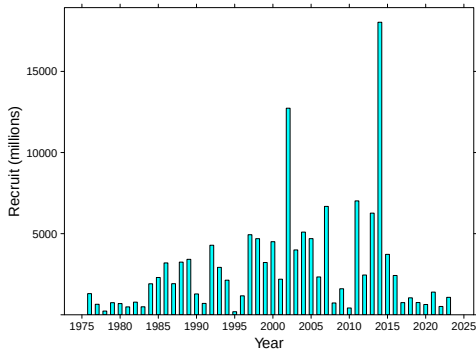


Mid-Atlantic Base Case: Estimated Biomass and Recruitment

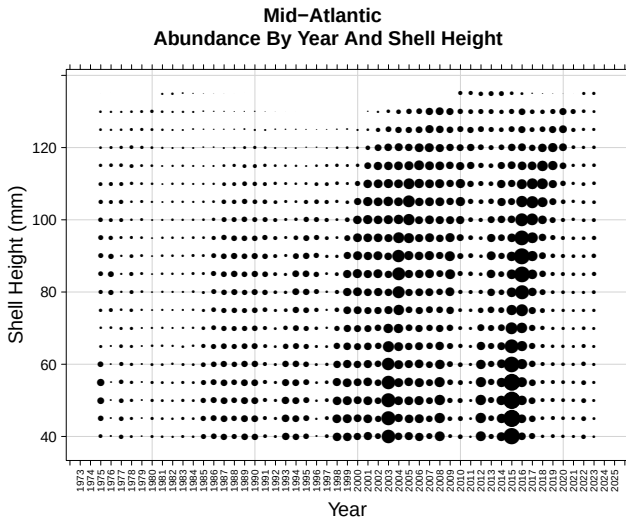
Mid-Atlantic
Survey And Model Estimated Biomass By Year



Mid-Atlantic
New Recruits By Year



Mid-Atlantic Base Case: Estimated Abundance by Size and Year



Symbol size is proportional to abundance

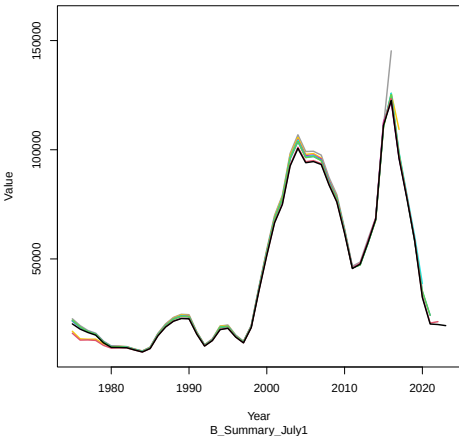
Mid-Atlantic Base Case: 2023 Fishing Mortality & Biomass

Stock	Biomass (mt meats)	CV	F	CV
Georges Bank Closed	28,883	0.08	0.469	0.34
Georges Bank Open	20,517	0.05	0.475	0.15
Mid-Atlantic	20,556	0.06	0.061	1.48
Total	69,956	0.04	0.33	0.05

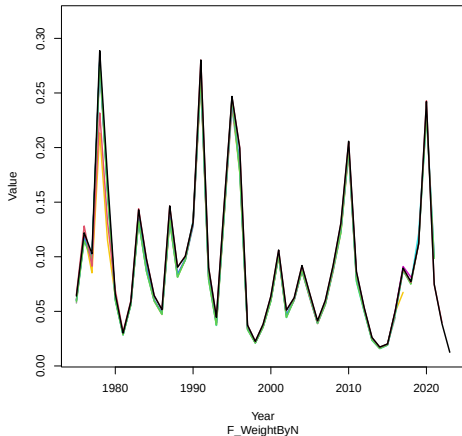
Scallops located in the deep water southeast portion of the Nantucket Lightship Area are excluded.

Mid-Atlantic Base Case: 7-Peel Retrospective Analysis

Retrospective Analysis Scallop CASA Model (peels=7, $\rho=0.14$)



Retrospective Analysis Scallop CASA Model (peels=7, $\rho=0.043$)

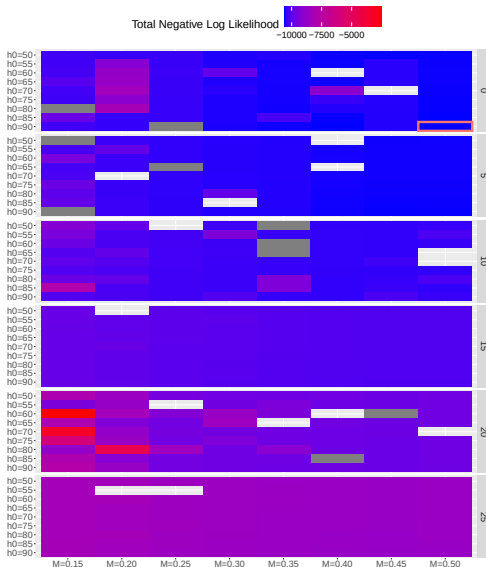


Mid-Atlantic Likelihood Profile Analysis

Following parameters were simultaneously profiled over a range of values:

- ▶ **model starting size:** 0mm to 25mm, by 5mm
- ▶ **mean M for juveniles and adults ($\bar{M}_{juv}$ and $\bar{M}_{ad}$):** 0.15 to 0.5, by 0.05
- ▶ **h_0 :** 50mm to 90mm, by 5mm

Results: Model favored starting at a smaller size, higher mean M , and a larger h_0 .



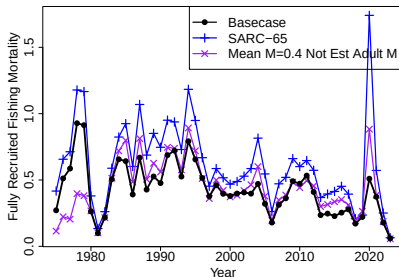
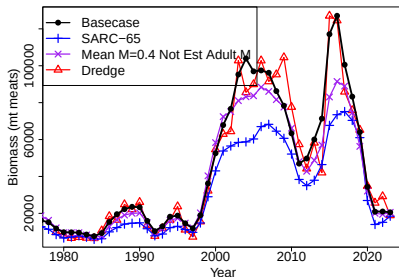
Mid-Atlantic Sensitivity Analysis

Following sensitivity runs were conducted:

- ▶ A bridge run using SARC-65 model configuration and data updated through 2023:
 $\bar{M}_{juv} = \bar{M}_{ad} = 0.25$, only estimated juvenile M .
- ▶ A run with $\bar{M}_{juv} = \bar{M}_{ad} = 0.4$, only estimated juvenile M .

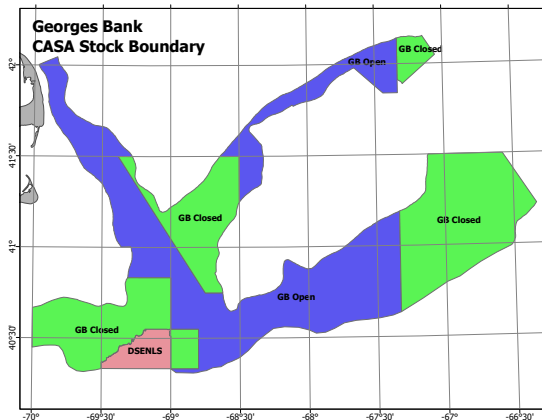
Results:

- ▶ Base case model outperformed the rest, comparing to dredge survey indices.
- ▶ Minimal effects on terminal years' estimates.



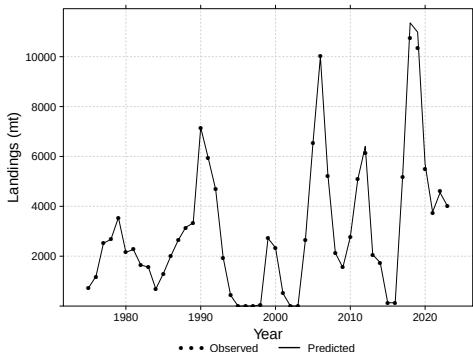
GB Closed Base Case Model Configuration

- Surveys: unlined dredge, lined dredge, SMAST large video camera, SMAST digital camera, and Habcam surveys.
- Fishery selectivity periods: 1975-1998, 1999-2016, 2017, 2018, 2019, 2020-2021, and 2022-2023.
- Assumed 2 growth periods, with the slowest growth in the 90s and recent years 2015-2023.
- M was estimated annually without varying by size.



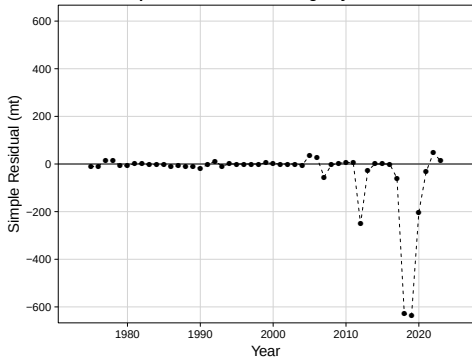
GB Closed Base Case: Observed/Predicted Catch

Georges Bank Closed Observed And Predicted Landings



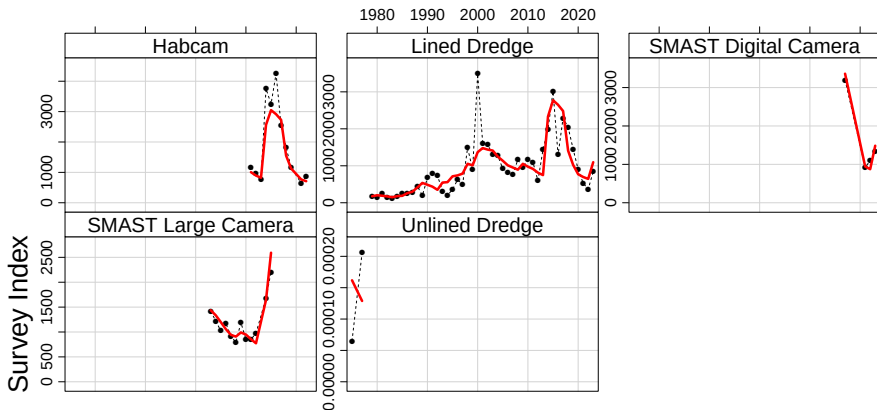
Georges Bank Closed

Simple Residuals Of Landings By Year



GB Closed Base Case: Observed/Predicted Survey Indices

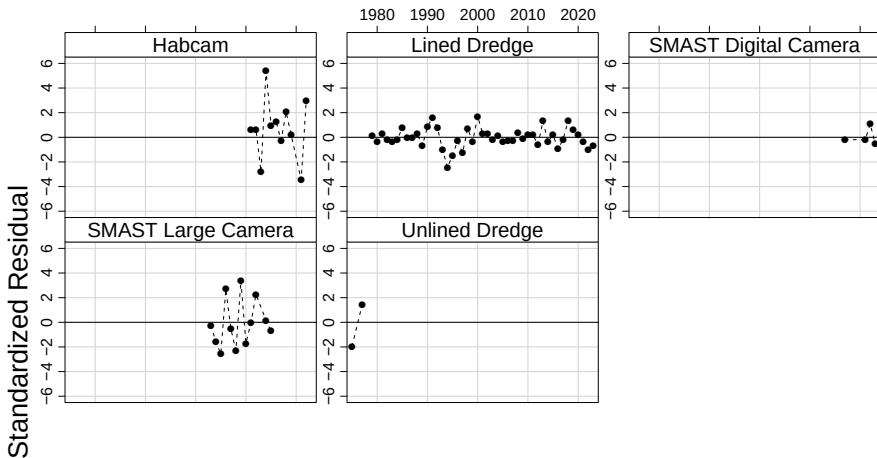
Georges Bank Closed Observed And Predicted Survey Indices By Year



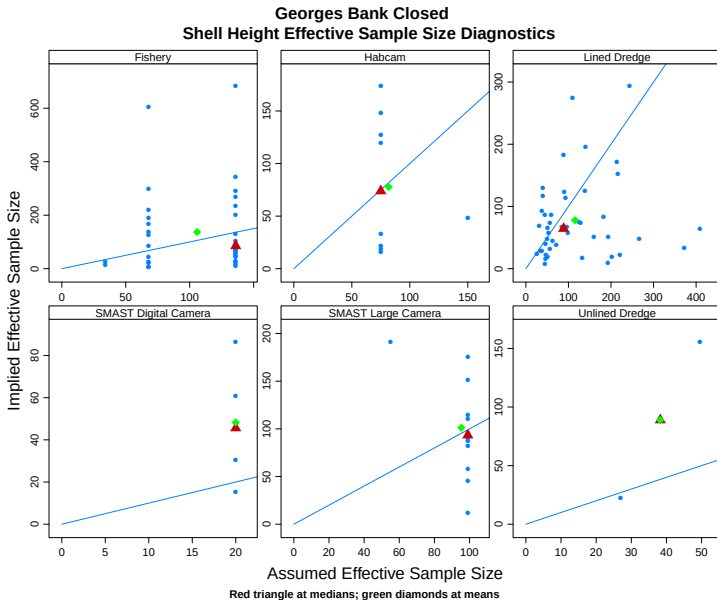
Black: Observed; Red: Predicted

GB Closed Base Case: Standardized Residuals of Survey Indices

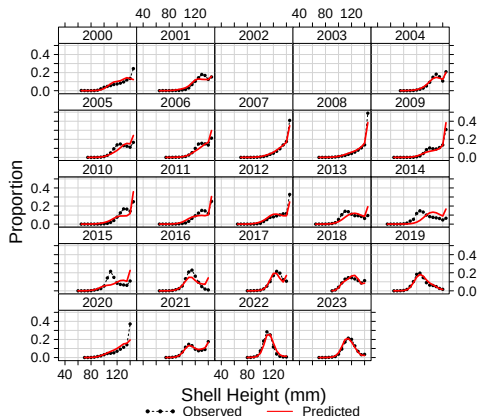
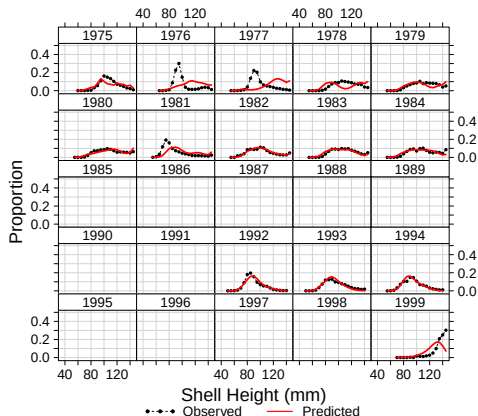
Georges Bank Closed Standardized Residuals Of Survey Indices By Year



GB Closed Base Case: Effective Sample Size

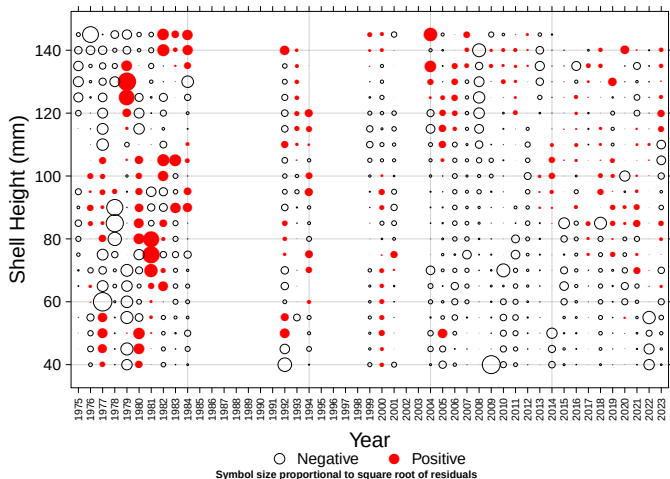


GB Closed Base Case: Fishery Size Freq.

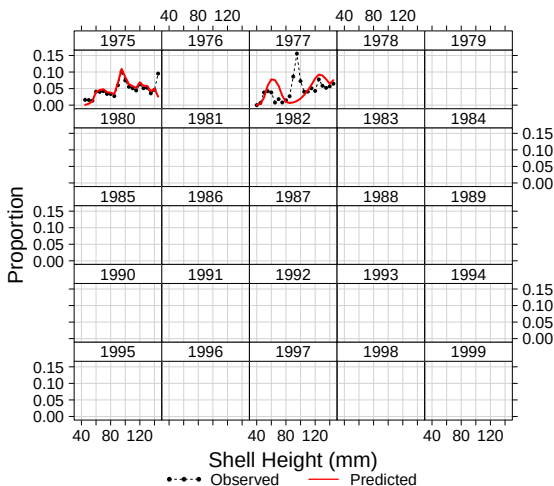


GB Closed Base Case: Fishery Size Freq. Residuals

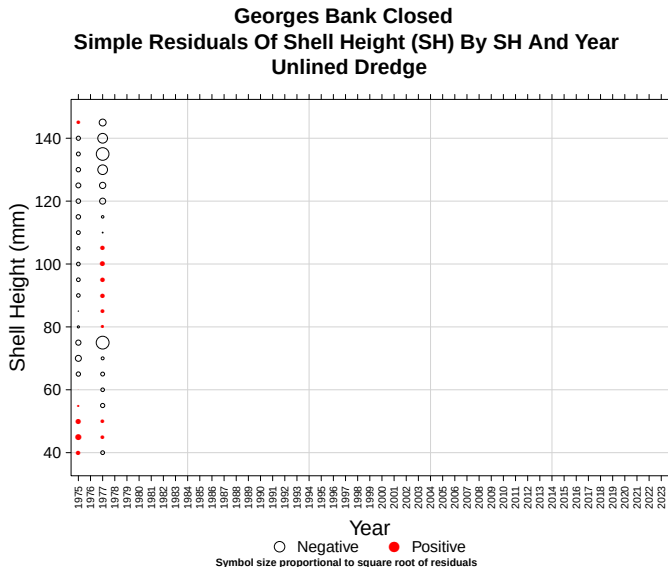
Georges Bank Closed
Simple Residuals Of Shell Height (SH) By SH And Year
Fishery



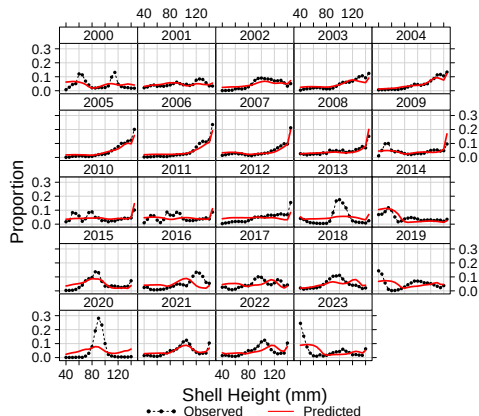
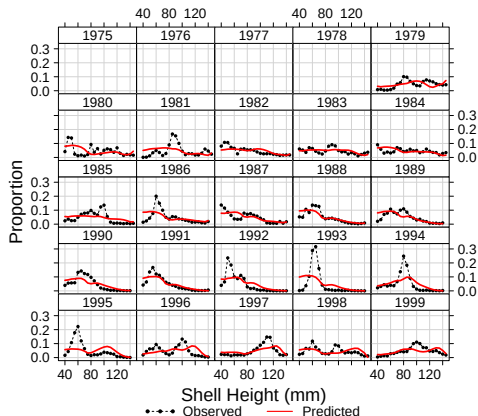
GB Closed Base Case: Unlined Dredge Size Freq.



GB Closed Base Case: Unlined Dredge Size Freq. Residuals

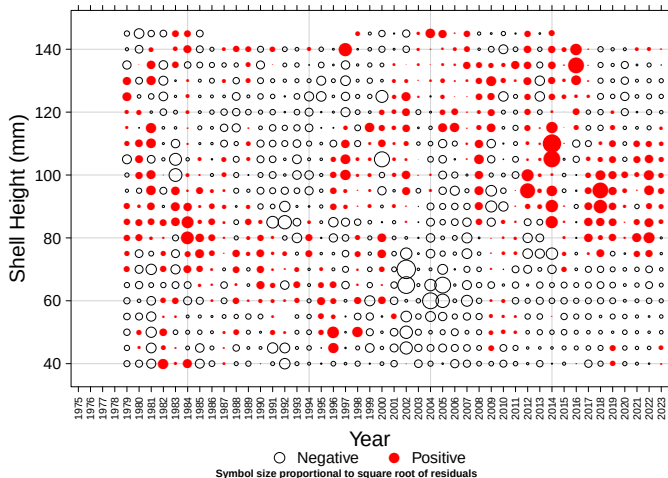


GB Closed Base Case: Lined Dredge Size Freq.



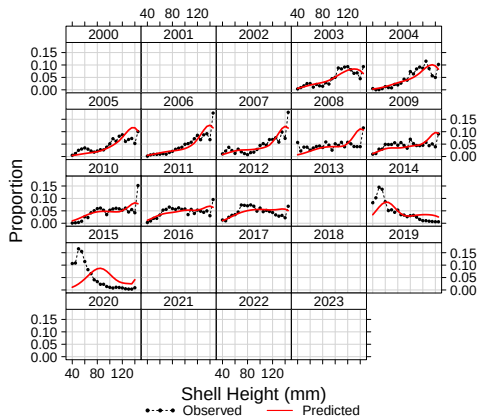
GB Closed Base Case: Lined Dredge Size Freq. Residuals

Georges Bank Closed
Simple Residuals Of Shell Height (SH) By SH And Year
Lined Dredge

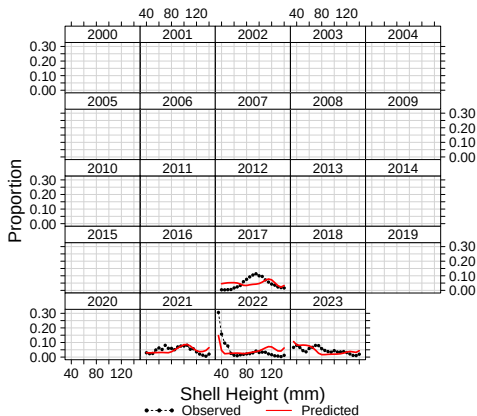


GB Closed Base Case: SMAST Size Freq.

SMAST Large Camera Survey

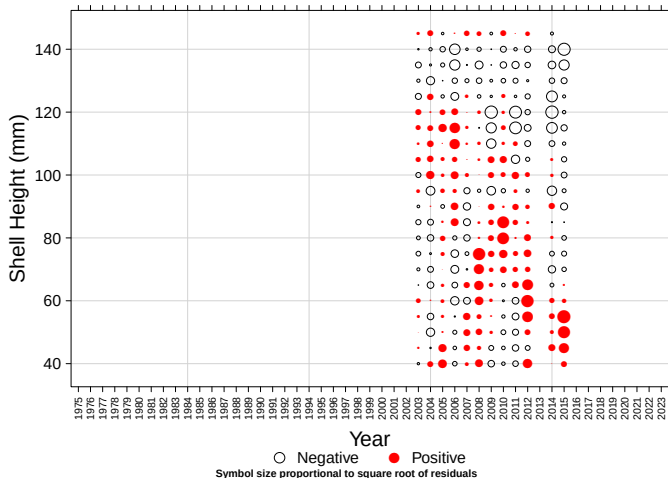


SMAST Digital Camera Survey



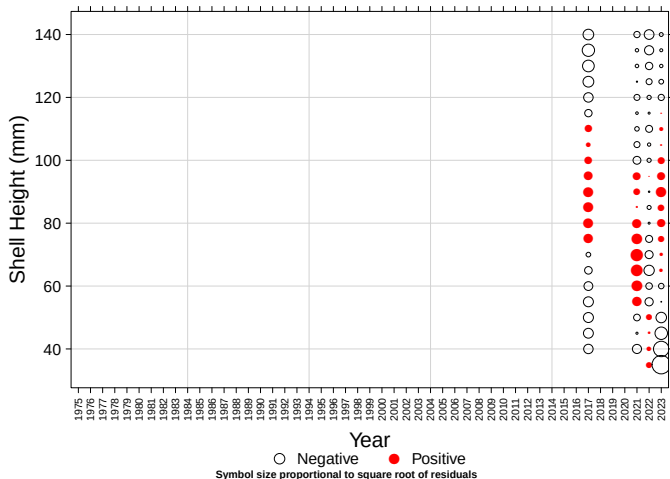
GB Closed Base Case: SMAST Large Camera Size Freq. Residuals

Georges Bank Closed
Simple Residuals Of Shell Height (SH) By SH And Year
SMAST Large Camera

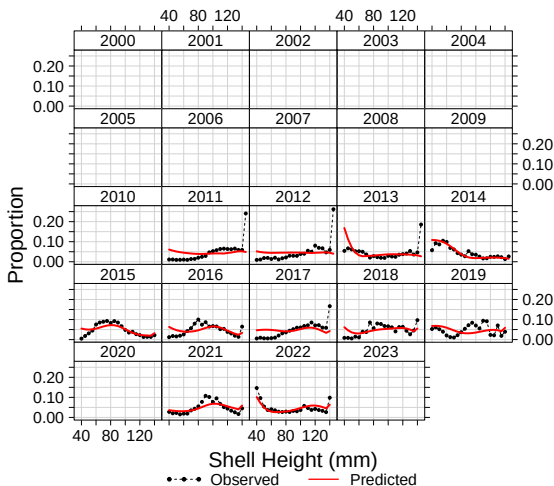


GB Closed Base Case: SMAST Digital Camera Size Freq. Residuals

Georges Bank Closed
Simple Residuals Of Shell Height (SH) By SH And Year
SMAST Digital Camera

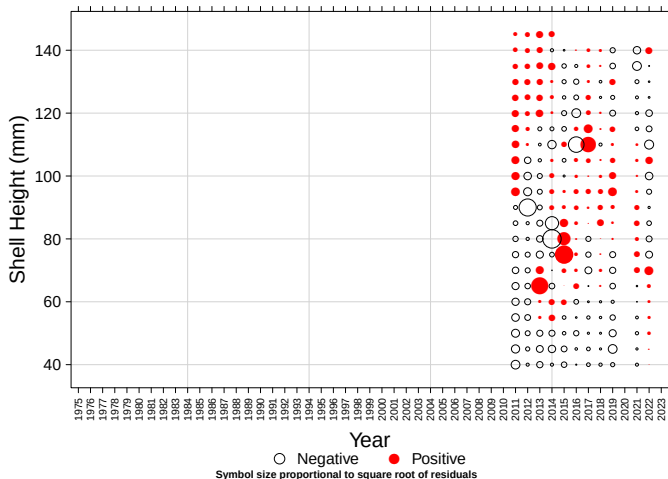


GB Closed Base Case: Habcam Survey Size Freq.

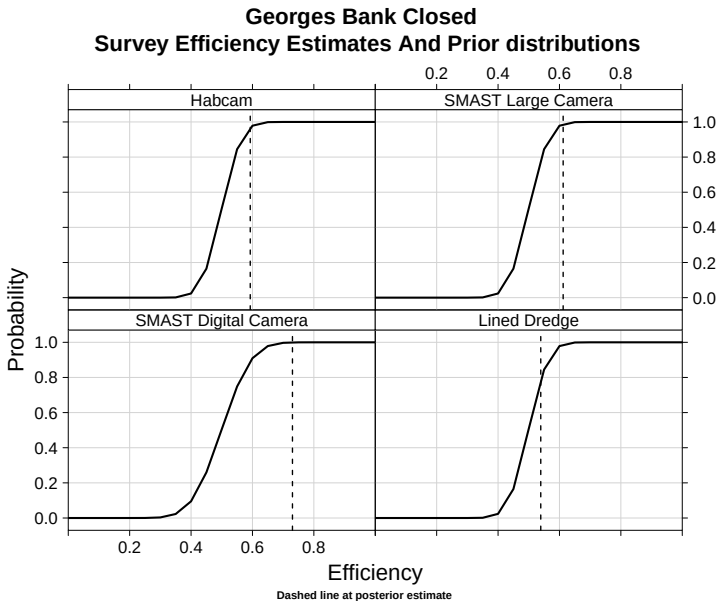


GB Closed Base Case: Habcam Size Freq. Residuals

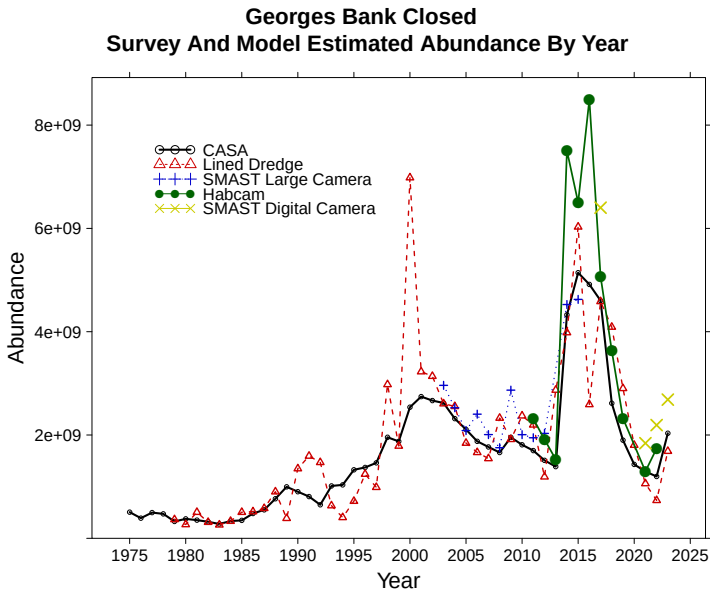
Georges Bank Closed
Simple Residuals Of Shell Height (SH) By SH And Year
Habcam



GB Closed Base Case: Survey Catchability

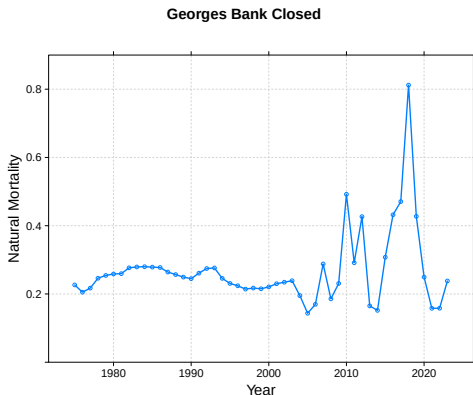


GB Closed Base Case: Observed and CASA-Estimated Abundance

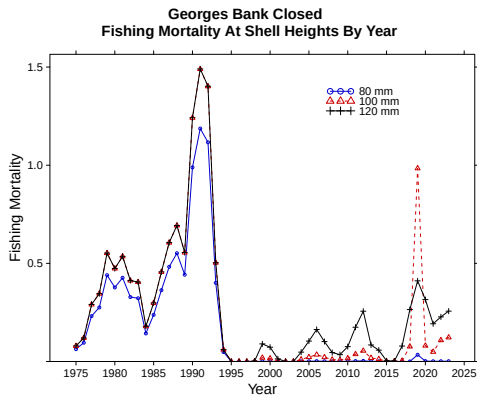
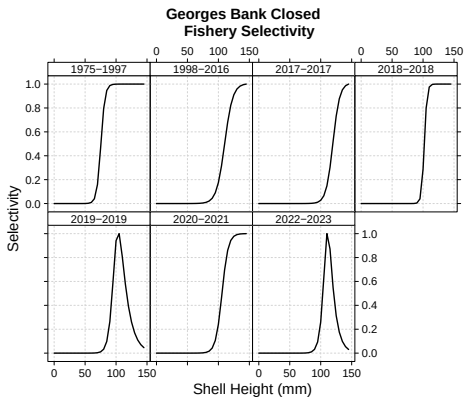


GB Closed Base Case: Natural Mortality By Year

- ▶ Mean M is 0.27 across all years and sizes for this assessment, while the mean M was 0.23 for SARC-65.
- ▶ M is more variable in recent years.
- ▶ Peak M was estimated at 0.81 for 2018, followed by 0.49 for 2010, coincide with gray meat disease and disappearance of scallops in the Lightship West.

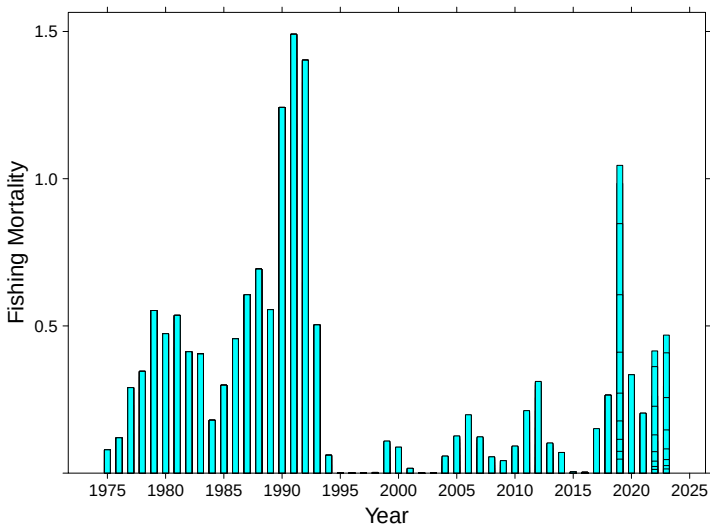


GB Closed Base Case: Fishing Selectivity and Mortality by Size and Year



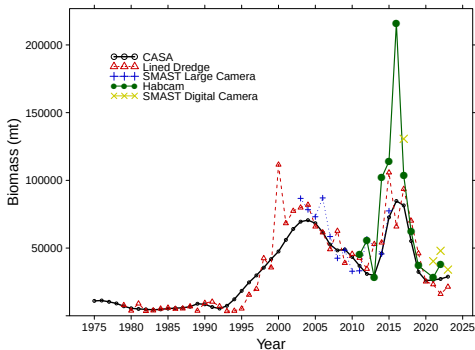
GB Closed Base Case: Estimated Fully Recruited Fishing Mortality

**Georges Bank Closed
Fully Recruited Fishing Mortality By Year**

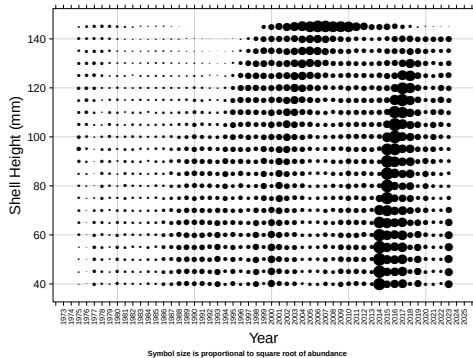


GB Closed Base Case: Estimated Biomass and Abundance

Georges Bank Closed
Survey And Model Estimated Biomass By Year

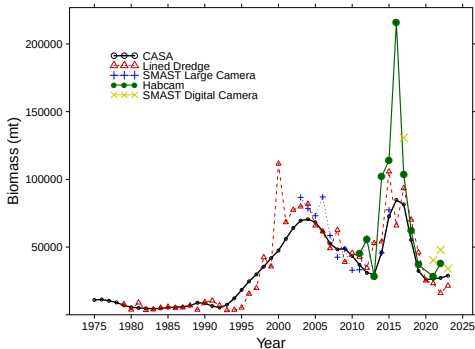


Georges Bank Closed
Abundance By Year And Shell Height

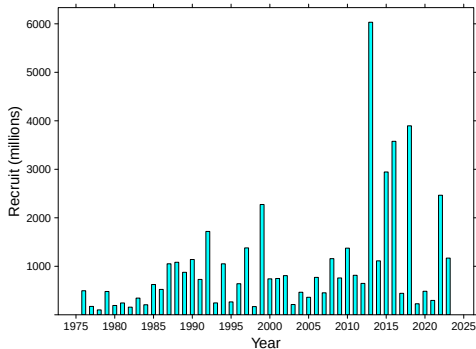


GB Closed Base Case: Estimated Biomass and Recruitment

Georges Bank Closed
Survey And Model Estimated Biomass By Year



Georges Bank Closed
New Recruits By Year



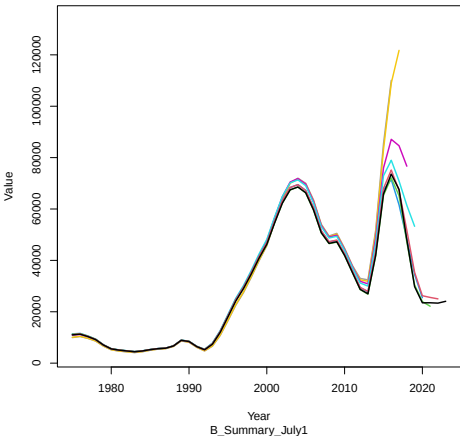
GB Closed Base Case: 2023 Fishing Mortality & Biomass

Stock	Biomass (mt meats)	CV	F	CV
Georges Bank Closed	28,883	0.08	0.469	0.34
Georges Bank Open	20,517	0.05	0.475	0.15
Mid-Atlantic	20,556	0.06	0.061	1.48
Total	69,956	0.04	0.33	0.05

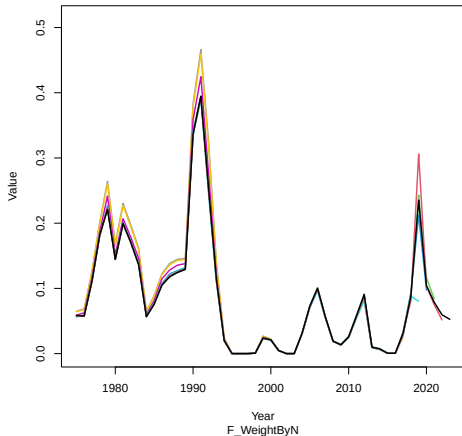
Scallops located in the deep water southeast portion of the Nantucket Lightship Area are excluded.

GB Closed Base Case: 7-Peel Retrospective Analysis

Retrospective Analysis Scallop CASA Model (peels=7, $\rho=0.4$)

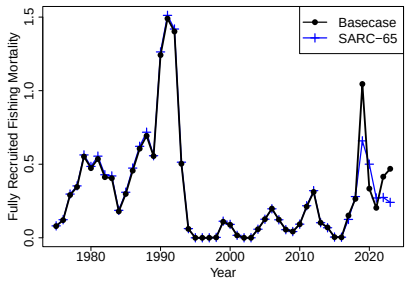
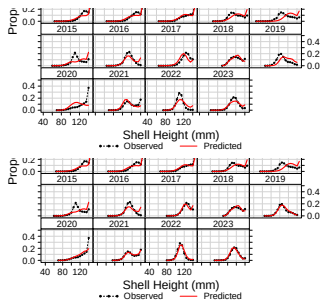
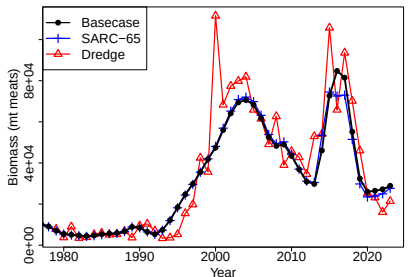


Retrospective Analysis Scallop CASA Model (peels=7, $\rho=-0.11$)



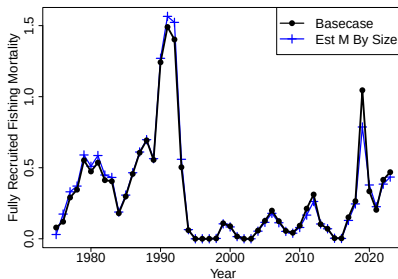
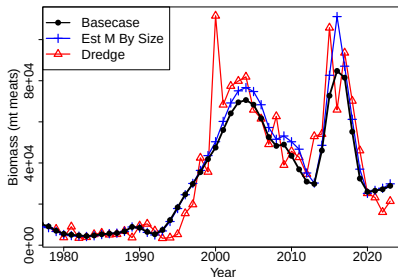
GB Closed Sensitivity Analysis - Bridge

- ▶ A bridge run using SARC-65 model configuration and data updated through 2023: fishery selectivity block for 2019 extended to 2023.
- ▶ Terminal year fishing mortality was about 35% lower than the base case model.



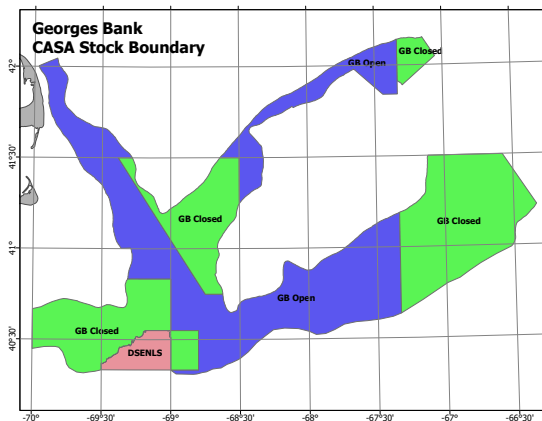
GB Closed Sensitivity Analysis - Natural Mortality By Size

- ▶ A run estimating M by size was conducted.
- ▶ Biomass estimates were slightly closer to the dredge survey compared to base case run.
- ▶ Estimated juvenile M for the first year was unrealistically high (over 3).



GB Open Base Case Model Configuration

- ▶ Surveys: unlined dredge, lined dredge, SMAST large video camera, SMAST digital camera, and Habcam surveys.
- ▶ Fishery selectivity periods: 1975-1998, 1999-2004, and 2005-2023.
- ▶ Assumed 2 growth periods, with the slowest growth in the 90s and recent years 2011-2023.



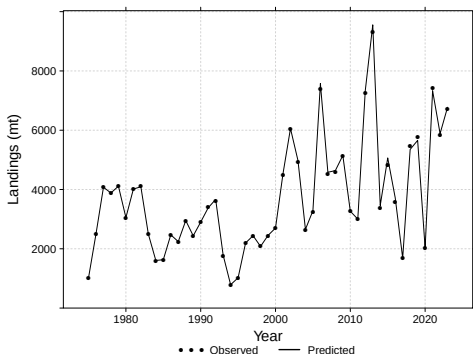
GB Open Base Case Model Configuration

Mean M was increased to 0.27 based on GB Closed estimates and $h_0 = 75\text{mm}$ based on Mid-Atlantic results.

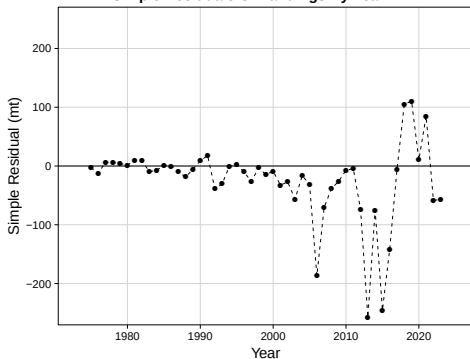
Natural Mortality Parameters	SARC-65	RT 2025
Option 1: $M(h, y) = \alpha(h)M_{\text{juv}}(y) + [1 - \alpha(h)]M_{\text{ad}}(y)$		
Mean Juvenile M [$\bar{M}_{\text{juv}}$]	Fixed at 0.2	Fixed at 0.27
Annual Deviation for Juvenile M [$\theta(y)$]	Estimated	Estimated
Mean Adult M [$\bar{M}_{\text{ad}}$]	Fixed at 0.2	Fixed at 0.27
Annual Deviation for Adult M [$\omega(y)$]	-	-
Logistic curve [$\alpha(h)$]	Fixed (65mm)	Fixed (75mm)
Option 2: $M(y) = \bar{M} + \gamma(y)$		
Mean Total M [$\bar{M}$]	-	-
Annual Deviation for Total M [$\gamma(y)$]	-	-

GB Open Base Case: Observed and Predicted Catch

Georges Bank Open Observed And Predicted Landings

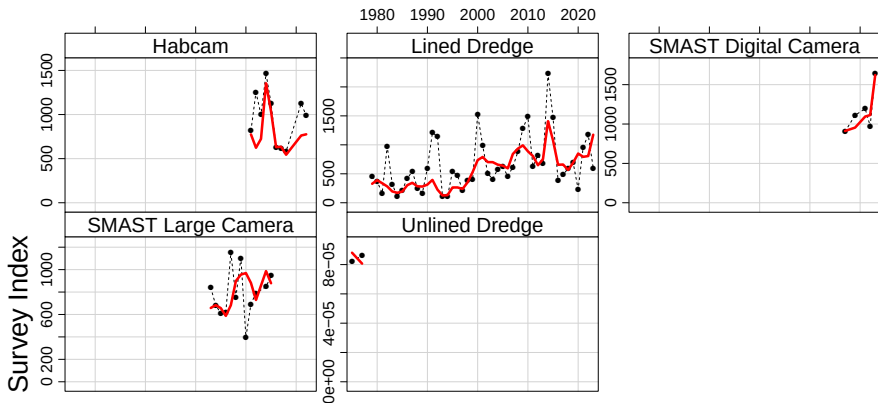


Georges Bank Open Simple Residuals Of Landings By Year



GB Open Base Case: Observed and Predicted Survey Indices

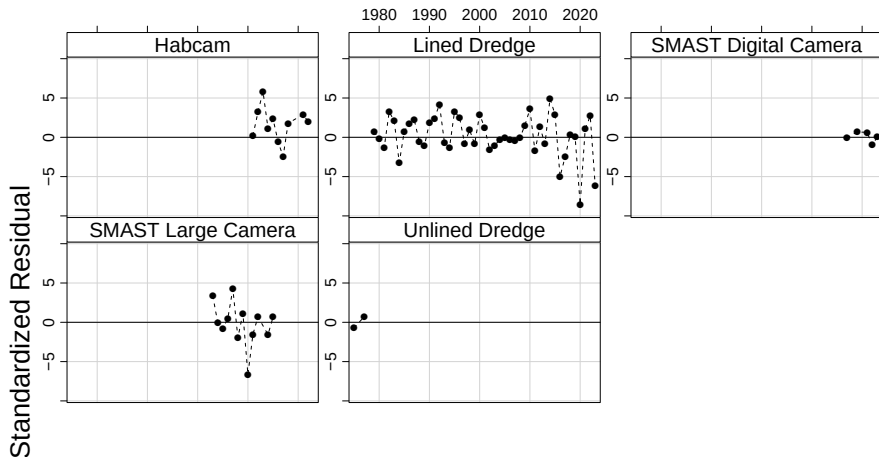
Georges Bank Open Observed And Predicted Survey Indices By Year



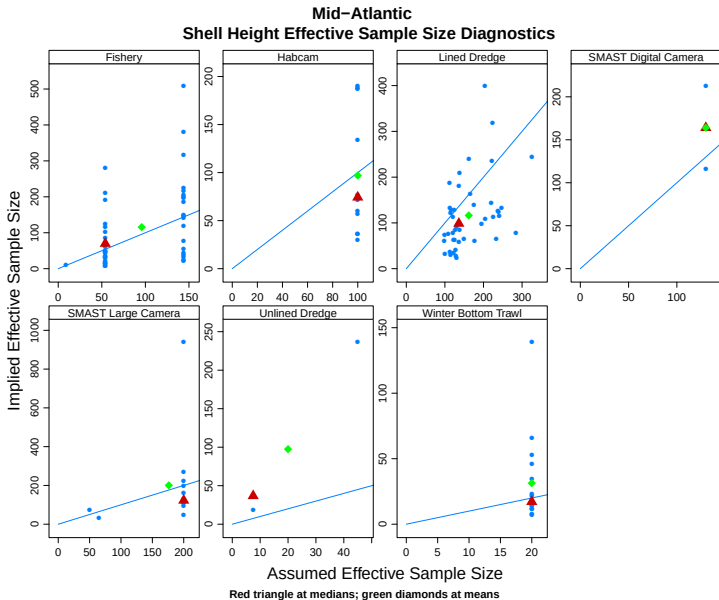
Black: Observed; Red: Predicted

GB Open Base Case: Standardized Residuals of Survey Indices

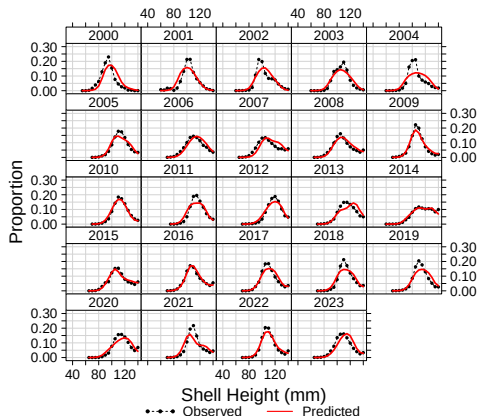
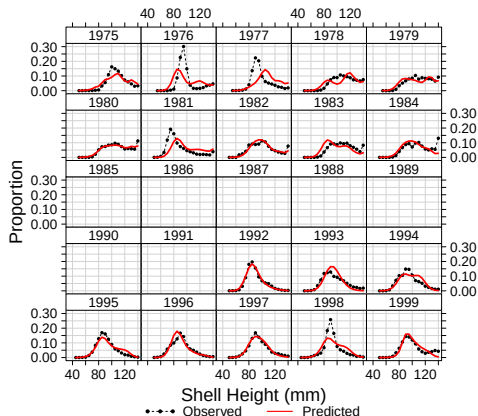
Georges Bank Open Standardized Residuals Of Survey Indices By Year



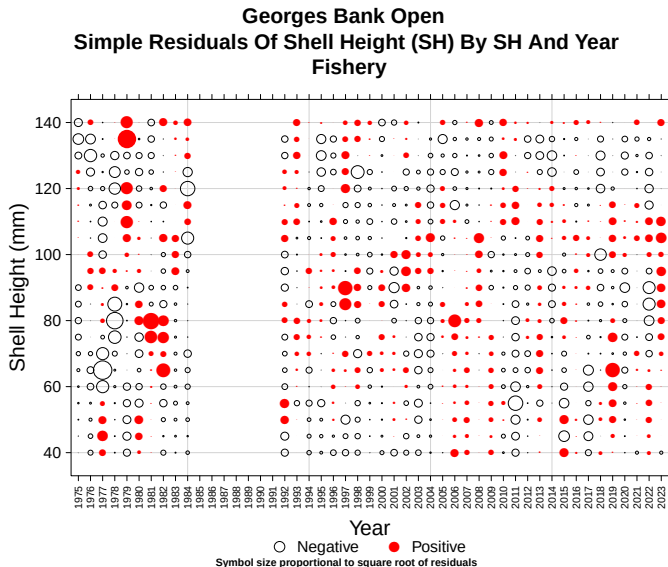
Mid-Atlantic Base Case: Effective Sample Size



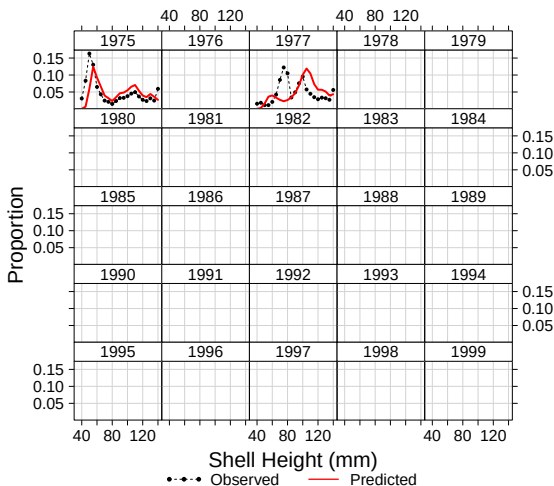
GB Open Base Case: Fishery Size Freq.



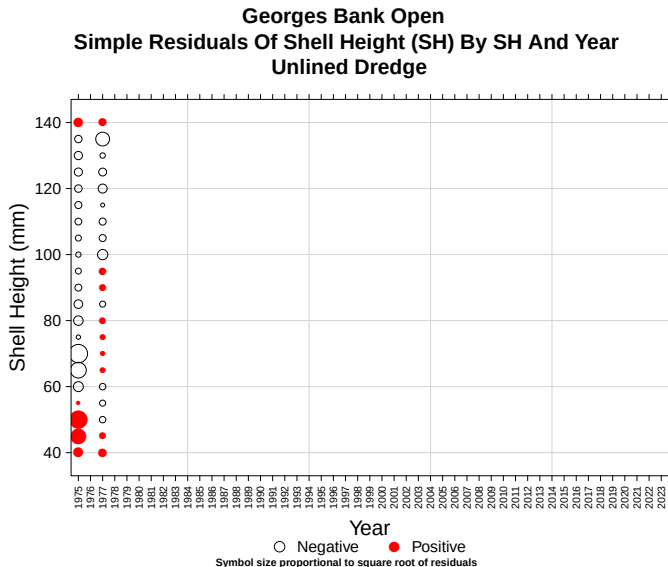
GB Open Base Case: Fishery Size Freq. Residuals



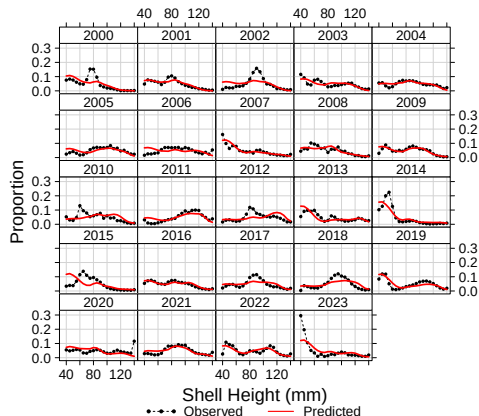
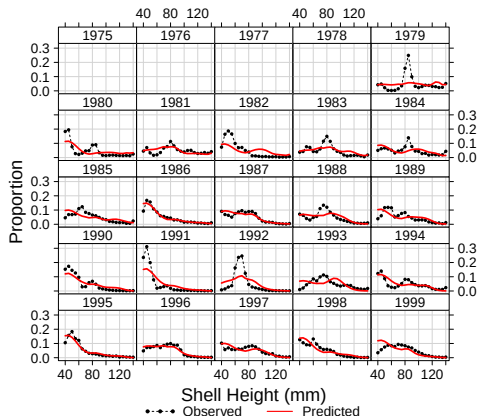
GB Open Base Case: Unlined Dredge Size Freq.



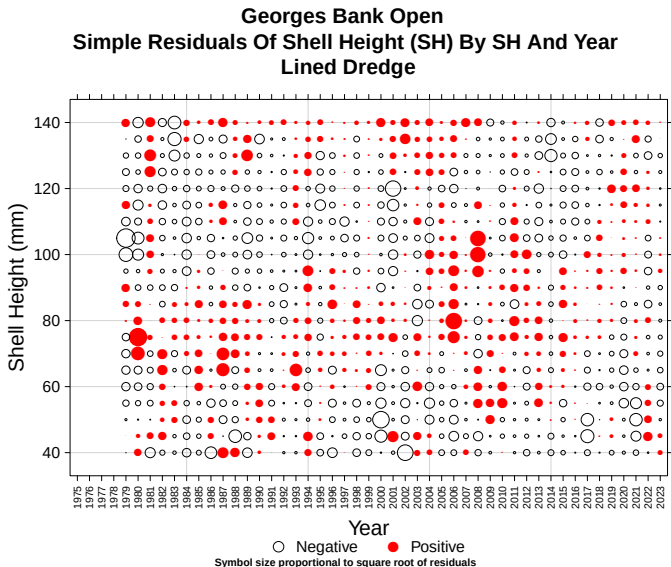
GB Open Base Case: Unlined Dredge Size Freq. Residuals



GB Open Base Case: Lined Dredge Size Freq.

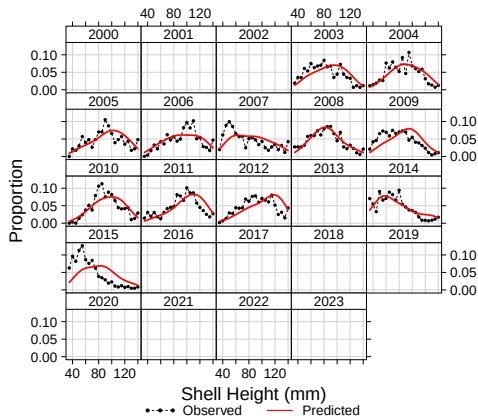


GB Open Base Case: Lined Dredge Size Freq. Residuals

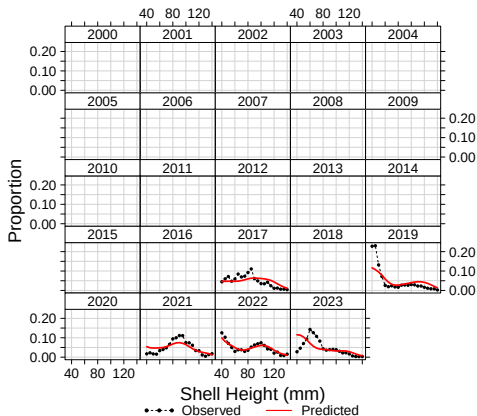


GB Open Base Case: SMAST Size Freq.

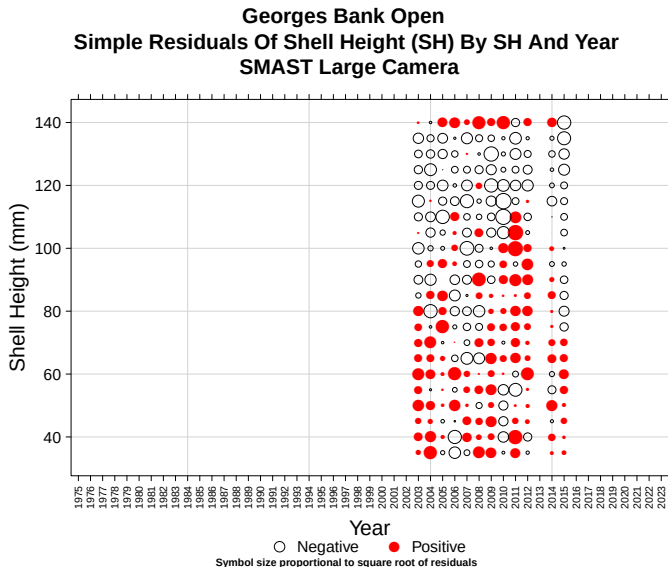
SMAST Large Camera Survey



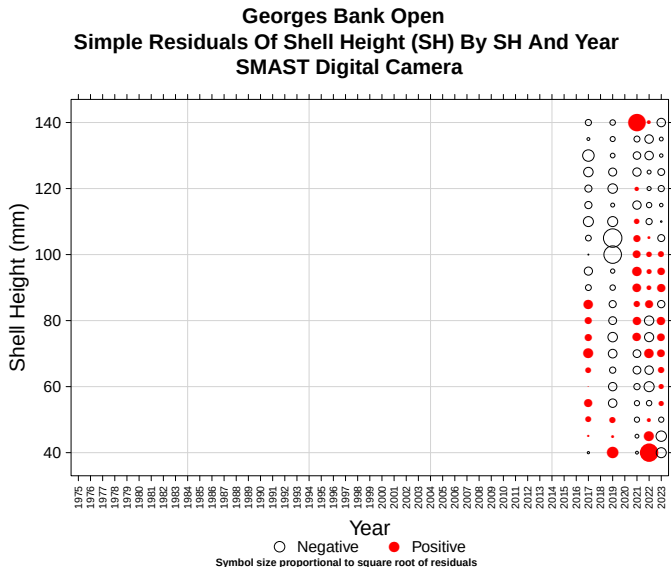
SMAST Digital Camera Survey



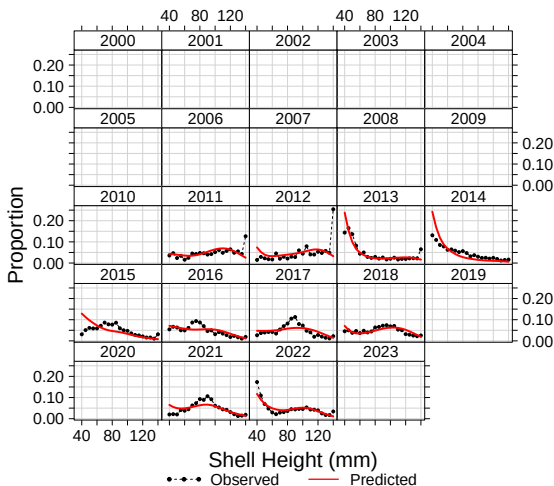
GB Open Base Case: SMAST Large Camera Size Freq. Residuals



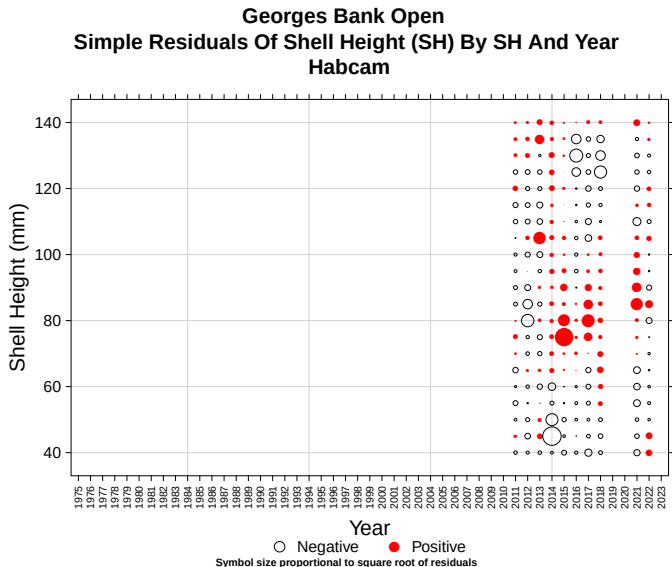
GB Open Base Case: SMAST Digital Camera Size Freq. Residuals



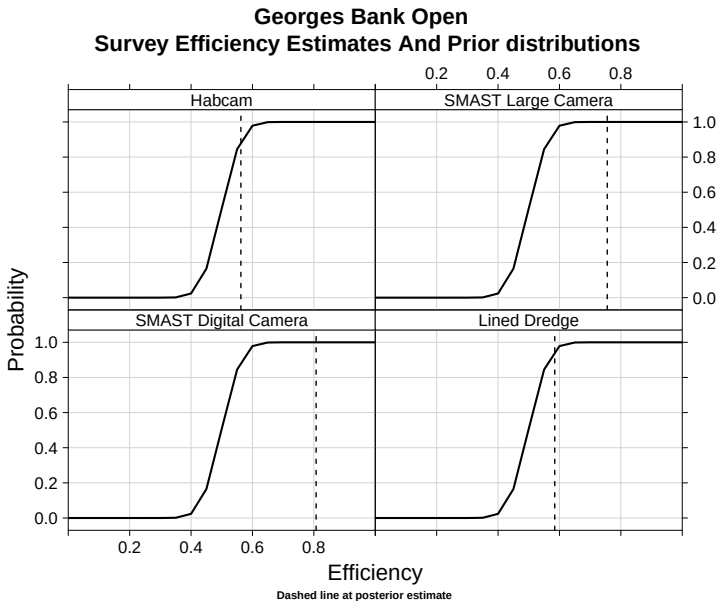
GB Open Base Case: Habcam Survey Size Freq.



GB Open Base Case: Habcam Size Freq. Residuals

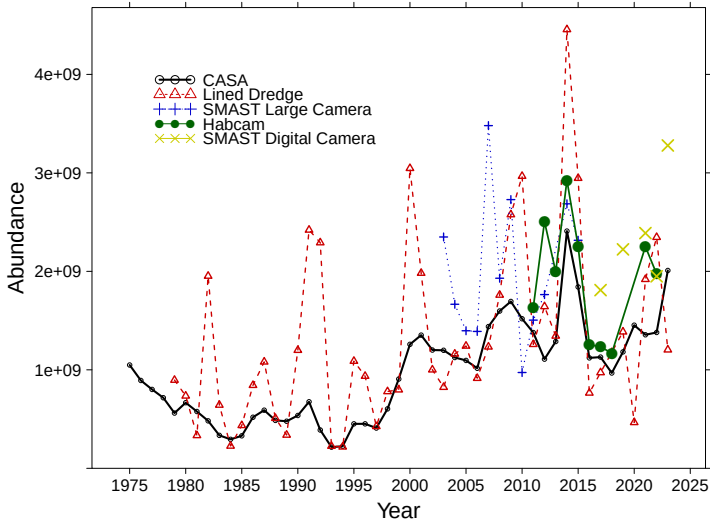


GB Open Base Case: Survey Catchability



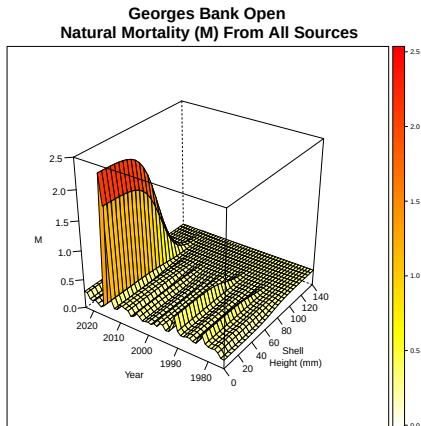
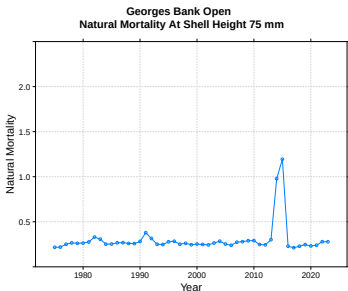
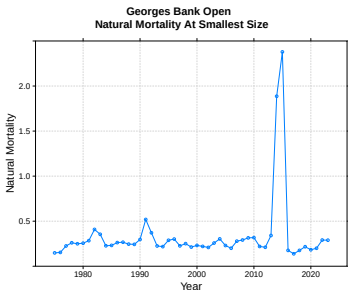
GB Open Base Case: Observed and CASA-Estimated Abundance

**Georges Bank Open
Survey And Model Estimated Abundance By Year**



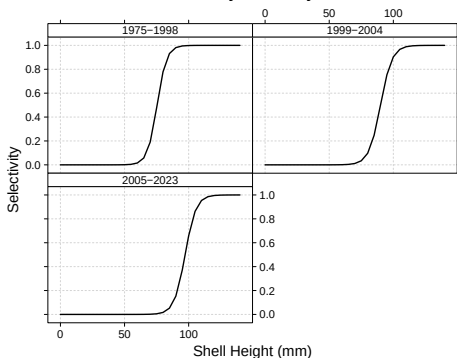
GB Open Base Case: Natural Mortality by Size and Year

- Mean M is 0.3 across all years and sizes for this assessment, while the mean M was 0.21 for SARC-65.
- Elevated juvenile M during large recruitment events.

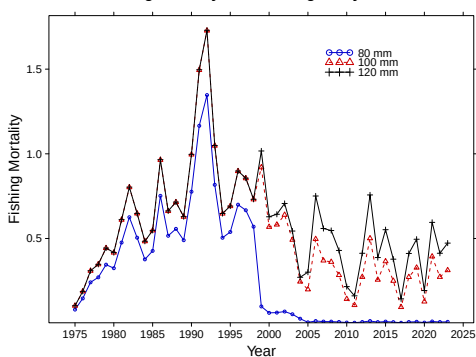


GB Open Base Case: Fishing Selectivity and Mortality by Size and Year

Georges Bank Open
Fishery Selectivity

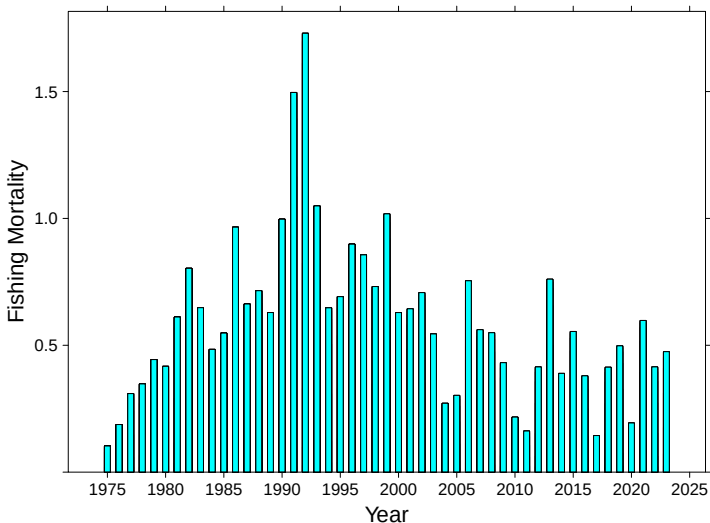


Georges Bank Open
Fishing Mortality At Shell Heights By Year

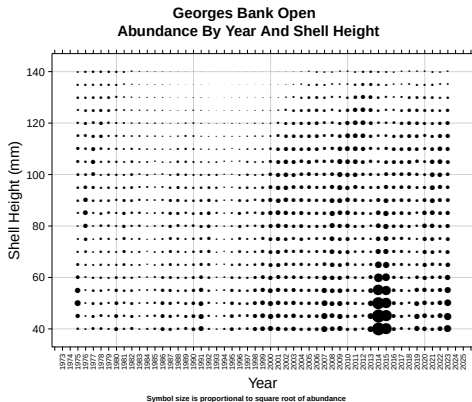
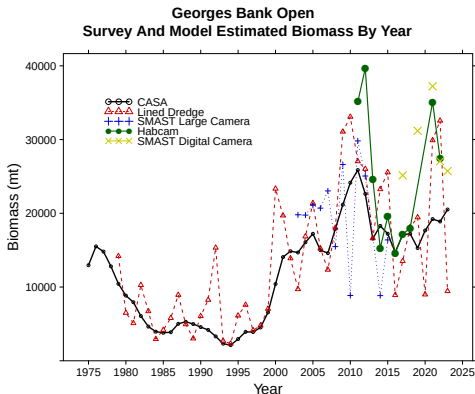


GB Open Base Case: Estimated Fully Recruited Fishing Mortality

**Georges Bank Open
Fully Recruited Fishing Mortality By Year**

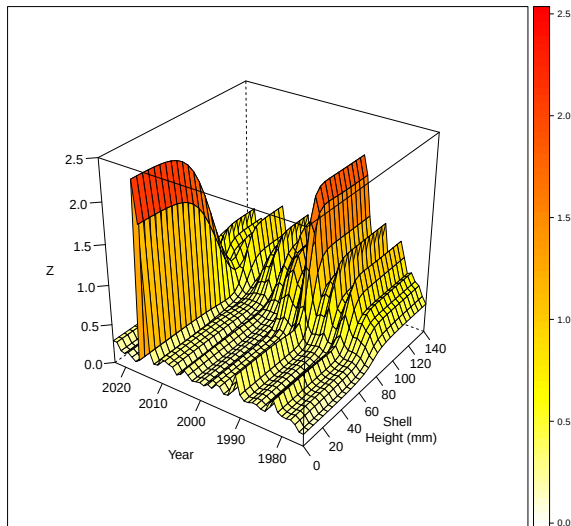


GB Open Base Case: Estimated Biomass and Abundance

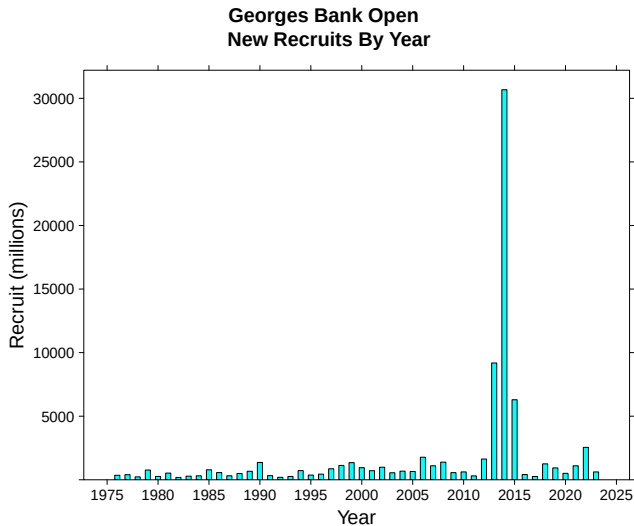


GB Open Base Case: Total Mortality

Georges Bank Open
Total Mortality (Z) From All Sources



GB Open Base Case: Estimated Recruitment



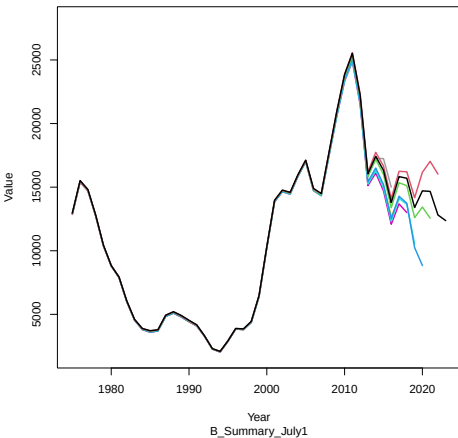
GB Open Base Case: 2023 Fishing Mortality & Biomass

Stock	Biomass (mt meats)	CV	F	CV
Georges Bank Closed	28,883	0.08	0.469	0.34
Georges Bank Open	20,517	0.05	0.475	0.15
Mid-Atlantic	20,556	0.06	0.061	1.48
Total	69,956	0.04	0.33	0.05

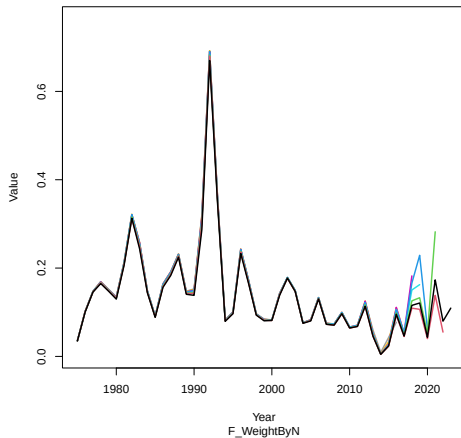
Scallops located in the deep water southeast portion of the Nantucket Lightship Area are excluded.

GB Open Base Case: 7-Peel Retrospective Analysis

Retrospective Analysis Scallop CASA Model (peels=7, $\rho = -0.1$)

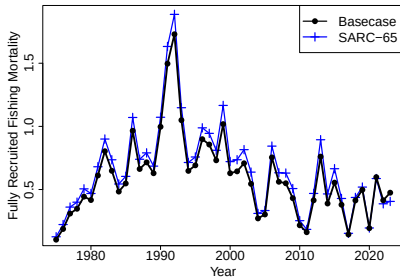
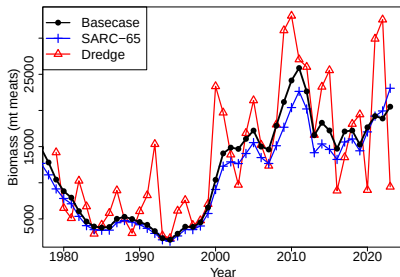


Retrospective Analysis Scallop CASA Model (peels=7, $\rho = 0.23$)



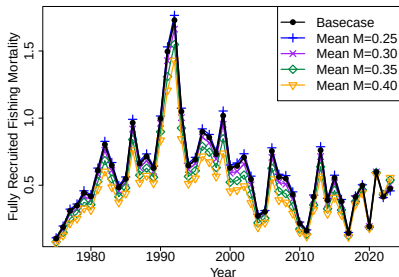
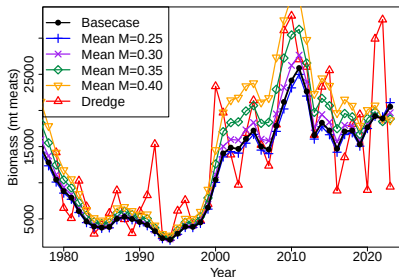
GB Open Sensitivity Analysis - Bridge

- ▶ A bridge run using SARC-65 model configuration and data updated through 2023: mean M fixed at 0.2.
- ▶ Base case model outperformed the Bridge, comparing to dredge survey indices.



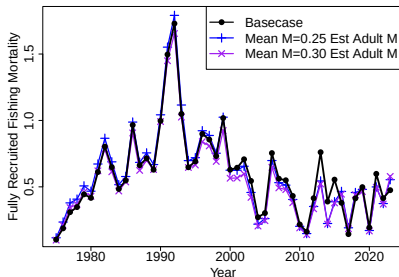
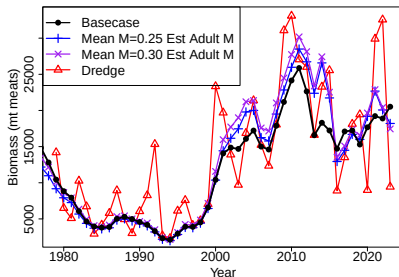
GB Open Sensitivity Analysis - Mean Natural Mortality

- Runs with various mean M (0.25 to 0.4 by 0.05) estimating M were conducted.
- Biomass estimates increased and fishing mortality decreased as mean M increased.
- Minimal effects on terminal years' estimates.

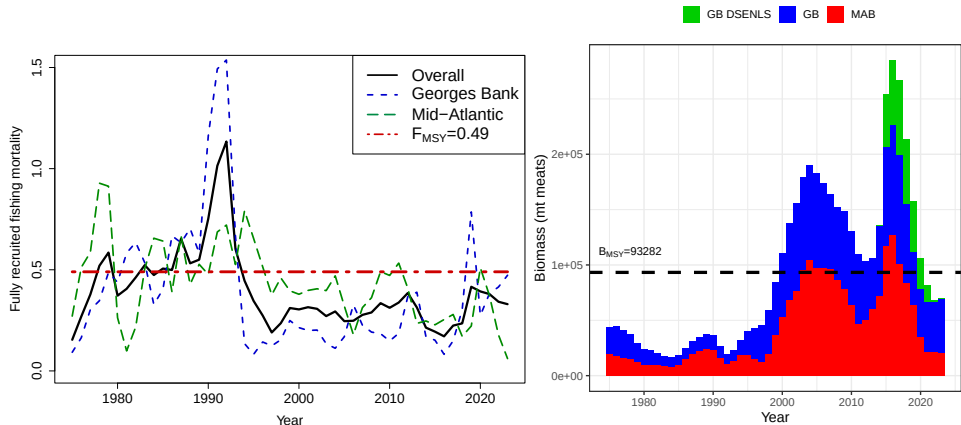


GB Open Sensitivity Analysis - Adult Natural Mortality

- Runs estimating adult M were conducted on two mean M values: 0.25 and 0.3.
- Biomass estimates were closer to the dredge survey compared to base case run.
- Estimated adult M was unrealistically high in 2015, higher than juvenile M .



Three Stocks Combined: Fishing Mortality & Biomass



Estimated fully recruited fishing mortality (left) and biomass (right) for Georges Bank (open and closed combined) and Mid-Atlantic sea scallops, including Habcam biomass estimates of scallops located in the deep water southeast portion of the Nantucket Lightship Area (DSENLS).

Three Stocks Combined: 2023 Fishing Mortality & Biomass

Stock	Biomass (mt meats)	CV	F	CV
Georges Bank Closed	28,883	0.08	0.469	0.34
Georges Bank Open	20,517	0.05	0.475	0.15
Mid-Atlantic	20,556	0.06	0.061	1.48
Total	69,956	0.04	0.33	0.05

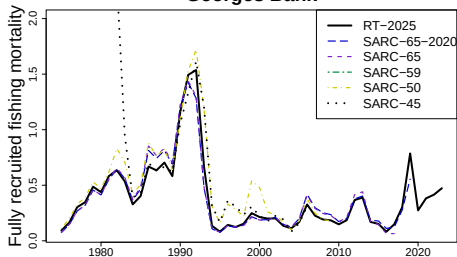
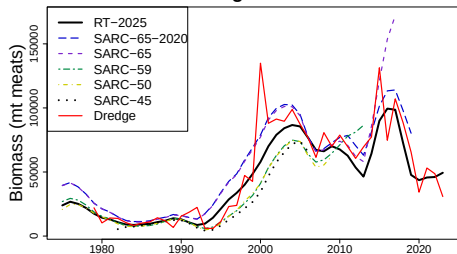
Scallops located in the deep water southeast portion of the Nantucket Lightship Area are excluded.

Historical Retrospective Analysis: SARC-45, 50, 59, 65, 65-2020

CASA Biomass & Fishing Mortality

Georges Bank

Georges Bank



Mid-Atlantic

Mid-Atlantic

